J.P.Morgan

Global Markets Strategy
05 August 2026

Base Metals Supply & Demand Tracker

Dislocation Derby

Copper:

- **US appetite for imports of refined copper continues to rage on.** The market is still awaiting any update following the US review of Section 232 copper tariffs with focus now on Trump's attendance this Friday at a State Department roundtable with mining executives. Nonetheless, the impact this lasting uncertainty is having on the market has been on full display over the last few weeks. In July, US bill of lading data suggest that the US imported more than 200 kt of refined copper last month, more than 80 kt higher than recent monthly levels and the highest import level for seaborne bill of lading inflows since data begin in 2014 **(Figure 35)**. Moreover, despite no update yet on refined copper cathode tariffs, the COMEX/LME arb still remains open for attracting imports into the US **(Figure 36).**
- **China comes off the boil for now...** Over the first half of July, China was showing classic signs of a tight domestic market. Low and drawing onshore inventories drove a push higher in Chinese premiums and a sustained opening in the SHFE/LME import arb which in turn prompted over 70 kt of cancelations from LME Asian warehouses. With relatively uneven demand across various sectors over the last few months **(Figure 28)**, the bout of tightness in China was primarily driven by a concentrated period of smelter maintenance, which weighed on Chinese refined supply from June **(Figure 45)**. As such, in the last few weeks, China's fundamental signals have cooled sharply as smelters have returned from this maintenance period. Inventory has stabilized more recently, albeit still at low levels **(Figure 33),** while China's import arb has swung closed, reflecting the ramp in domestic supply and imports from the LME on their way **(Figure 38)**.
- **But how long can it stay there?** With both the US and China pulling on the market for much of July, LME on-warrant stocks have now fallen below 100 kt, a level of low on-warrant inventory where backwardation can ramp up aggressively and LME 3M copper prices enter a more asymmetric bullish pricing regime **(Figure 1 & Figure 2)**. In the near-term, if LME backwardation further flares out, we will be watching for the potential for another sustained opening in China's export arb for copper, which would be followed by a spate of opportunistic exports to backfill LME warehouses once again. However, as long as the US import arb remains open, this is only a temporary fix. Concentrate is scarce—Chinese imports of copper concentrate are down by -1% ytd yoy in 6M26 **(Figure 40)**—and negative and falling treatment charges are reflecting this as mine supply continues to stumble **(Figure 39)**. Moreover, China is sitting on low inventory at the moment ahead of a second seasonal peak in demand this autumn, likely coinciding with accelerating budget deployment and fiscal easing on the macro front. Overall, this all likely means while there will be a give and take at times, as long as the US import arb remains open (e.g. escalating tariff expectations re-enforced), China will continue to need to chase prices higher to ultimately secure its needed copper units. **As we see this tug of war continuing, we expect copper prices will likely persistently see higher floors and higher ceilings on a push towards**

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See page 31 for analyst certification and important disclosures.

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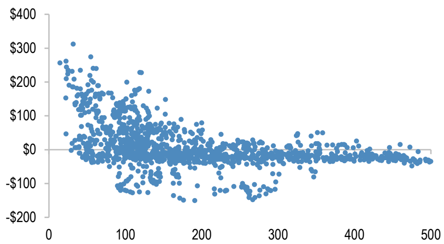
$15,000/mt over 2H26, with risks still skewed towards reaching a very bullish pinch point driven by rapidly dwindling ex-US inventories.

Figure 1: LME on-warrant copper stocks vs LME cash-3M spread, weekly

X-axis: Thousand mt, Y-axis: US$/mt (+ = backwardation / - = contango)



Data since 2000, x-axis truncated to 500 kmt
Source: LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 2: Weekly returns of LME 3M copper prices under different LME on-warrant inventory environments

On-warrant LME inventory of...		Positive Hit Rate	Average Weekly Return	Median Weekly Return
100 kmt or less and...	Decreasing WoW	63%	0.96%	1.09%
	Increasing WoW	50%	0.14%	0.01%
	Either	57%	0.56%	0.64%
Greater than 100 kmt and...	Decreasing WoW	58%	0.61%	0.39%
	Increasing WoW	45%	-0.42%	-0.47%
	Either	51%	0.11%	0.07%

Weekly data since 2000
Source: LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Aluminium:

- **LME prices still likely need to incentivize boosted Chinese semi exports to bridge the 3Q26 RoW supply gap even as Middle East supply gradually ramps up.** Coinciding with a consistent and robust draw in onshore Chinese aluminum ingot inventories **(Figure 58)**, exports of Chinese semi-fabricated aluminum products ramped up to over 700 kt in June, +45% yoy **(Figure 62)**. That being said, China's semi-fabricated export arb, needed to pull out these boosted level of swing exports, has now been closed since mid-June, risking an unwind in this elevated export flow from this month onwards **(Figure 65)**. While Middle East producers continue to make progress restoring supply, our balances still show a meaningful supply shortfall this quarter before greater levels of oversupply arrive over the coming year. Hence, we see a further draw in inventory needed this quarter with the vast majority of visible aluminum inventory still sitting in China at the moment **(Figure 56)**. Piecing it all together, while looser aluminum balances are coming eventually, we still see a sizeable 3Q26 ex-China deficit that will likely need to continue to pull on China's inventory this quarter, skewing risk towards another push higher in LME prices/backwardation to once again re-open the export arb and bridge the supply gap until we reach a more well-supplied global market once again next year.

Zinc:

- **Long-awaited opening of China's export arb is finally upon us.** For the last few months, focus in zinc has been on the bifurcated nature of balances in China (loose with onshore social stocks now over 270 kt **(Figure 79)**) vs ex-China (tight) with the question of when (more than if) China's export arb would open to alleviate this dislocation. Drawing on-warrant LME inventory of zinc over the last month has coincided with a steepening backwardation at the front of the LME zinc curve which is finally beginning to more materially jerk open the export arb **(Figure 88)**. LME prices and backwardation could still have further to run in the next few weeks to secure the pull of this metal out of China, but eventually we will see a growing export flow in response, similar to 4Q25 **(Figure 87)**. Hence, on the other end of this tug by LME prices, we eventually see China's zinc surplus helping to relieve tightness in the rest of world, unwinding some of zinc's premium and pressuring LME prices back towards a $3,400-3,500/mt range as we move towards 4Q26.

Nickel:

- **Indonesian HPAL producers still face cost pressure**. Our equity analysts estimate sulphur costs alone still account for $10k/t of Indonesian HPAL producers' cash costs, vs only about $3-4k/t a year ago. They also flag that battery demand is unlikely to match the pace of planned HPAL project additions which, in their view, increases the risk that some projects see delays or cancelations. In the near-term, all eyes remain on government policy around any increase in mining quotas for 2H26. As of now, the only guidance from the government has been an emphasis that additions in mining quotas will be limited to meet demand, rather than a broader flood like in 2025. Our analysts on the ground now expect that Indonesia will likely increase its ore quota by ~40-50mn wmt, reaching 300-310mn wmt, vs a total ore demand of 330 mn wmt, with the remaining 20 mn wmt being met by imports from the Philippines. From a high level, we still expect the Indonesian RKAB policy will remain active and flexible over the coming years to attempt to maintain balance and continue to defend and support a desired price outcome (not too high, not too low) and maximize tax revenue.

Regular publication of JP Morgan Metals Research notes will recommence in the second half of August. We wish all our readers a great end of summer.

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Global demand indicators

Table 1: Global metals monthly price change heatmap

Percent change, MoM & YoY

	Global Metals Price Heatmap	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	2023	2024	2025	2026 YTD
Metals Prices	LME Copper	3.9%	-2.6%	3.0%	3.7%	6.0%	2.8%	11.0%	5.9%	1.4%	-7.6%	5.3%	5.0%	-1.9%	3.1%	2.2%	2.4%	41.7%	11.6%
	LME Aluminium	6.3%	-1.3%	2.0%	2.5%	7.6%	-0.6%	4.4%	5.0%	-0.1%	10.4%	0.2%	5.5%	-15.8%	3.2%	0.3%	7.0%	17.4%	7.3%
	LME Zinc	5.0%	0.3%	2.1%	5.0%	3.2%	0.0%	2.0%	9.1%	-2.5%	-2.7%	4.2%	5.3%	0.3%	2.6%	-10.6%	12.1%	4.7%	18.2%
	LME Nickel	-0.1%	-1.8%	3.2%	-1.2%	-0.1%	-2.6%	12.3%	7.9%	-0.6%	-4.1%	13.8%	-2.1%	-14.6%	5.9%	-44.7%	-7.7%	8.6%	2.0%
	LME Lead	4.4%	-3.6%	1.0%	-0.1%	1.4%	-1.8%	1.5%	-0.1%	-2.3%	-3.0%	2.8%	3.1%	-7.0%	0.1%	-9.8%	-5.6%	3.0%	-6.7%
	LME Tin	10.9%	-3.0%	7.1%	1.1%	1.9%	8.5%	3.6%	28.1%	11.1%	-19.0%	5.3%	12.6%	-6.9%	7.2%	2.4%	14.4%	39.4%	35.9%
	LME Cobalt spot	-1.1%	0.0%	0.0%	5.0%	38.8%	0.0%	9.9%	5.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	-43.9%	-16.6%	119.6%	5.6%
	LME Lithium Hydroxide	-3.2%	-2.1%	5.4%	11.9%	1.0%	5.8%	10.1%	62.0%	-4.3%	13.8%	3.5%	7.8%	-8.3%	-5.6%	-80.4%	-42.9%	18.5%	70.5%
	COMEX Copper	7.5%	-13.4%	3.8%	7.5%	4.8%	1.9%	9.6%	4.3%	1.4%	-6.5%	5.6%	7.8%	-3.1%	4.4%	2.1%	3.5%	41.1%	15.1%
	SHFE Copper	2.4%	-1.9%	1.0%	5.1%	4.8%	0.0%	13.1%	8.3%	-4.2%	-6.6%	5.8%	3.6%	-2.3%	3.3%	4.4%	6.8%	33.7%	7.4%
	SHFE Aluminium	2.8%	-0.5%	0.6%	0.0%	2.5%	1.1%	5.6%	10.6%	-6.0%	4.6%	-1.3%	-0.5%	-7.2%	5.0%	4.5%	1.0%	14.7%	3.8%
	SHFE Zinc	-0.5%	-0.6%	-1.3%	-1.0%	2.0%	0.2%	4.1%	12.2%	-6.0%	-4.1%	0.4%	5.1%	-2.5%	2.8%	-9.5%	19.0%	-9.4%	7.8%
	Spot Gold	0.4%	-0.4%	4.8%	11.9%	3.7%	5.9%	1.9%	13.3%	7.9%	-11.6%	-1.1%	-1.7%	-11.7%	1.0%	13.1%	27.2%	64.6%	-6.0%
	Spot Silver	9.5%	1.7%	8.2%	17.4%	4.4%	16.0%	26.8%	18.9%	10.1%	-19.9%	-1.9%	2.1%	-22.2%	-1.7%	-0.7%	21.5%	148.0%	-18.6%
	Spot Platinum	28.5%	-5.0%	6.1%	14.9%	-0.1%	6.1%	23.3%	6.5%	7.9%	-17.5%	1.7%	-3.4%	-19.1%	6.2%	-7.7%	-8.5%	127.0%	-19.5%
	Spot Palladium	13.6%	8.3%	-7.9%	14.3%	14.1%	1.1%	11.4%	5.7%	4.4%	-17.2%	3.6%	-11.2%	-10.9%	5.9%	-38.6%	-17.1%	77.5%	-20.7%

*LME is 3-month rolling forward. Lithium hydroxide and SHFE prices are futures contracts. Prices as of 30th July 2026.
Source: Bloomberg Finance L.P.

Table 2: Global manufacturing PMI by country

		Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26
Groups	**JP Morgan Global**	50.3	49.8	49.5	50.4	49.7	50.9	50.7	50.7	50.5	50.4	50.9	51.8	51.3	52.6	52.7	52.2
	Developed Markets	49.1	49.1	50.0	50.5	49.2	50.9	50.3	50.3	50.5	50.5	51.4	51.8	52.0	53.8	53.9	53.2
	Emerging Markets	51.3	50.5	49.2	50.4	50.1	50.9	51.2	51.2	50.5	50.4	50.6	52.0	50.7	51.6	51.6	51.4
	European Union	48.6	49.1	49.2	49.2	49.6	50.4	49.7	49.7	49.5	48.8	49.5	50.5	51.3	52.0	51.4	51.2
	Eurozone	48.6	49.0	49.4	49.5	49.8	50.7	49.8	49.8	49.6	48.8	49.5	50.8	51.6	52.2	51.6	51.4
Asia Pacific	China (Caixin/Markit)	51.2	50.4	48.3	50.4	49.5	50.5	51.2	51.2	49.9	50.1	50.3	52.1	50.8	52.2	51.8	51.7
	China (Official PMI)	50.5	49.0	49.5	49.7	49.3	49.4	49.8	49.8	49.2	50.1	49.3	49.0	50.4	50.3	50.0	50.3
	India	58.1	58.2	57.6	58.4	59.1	59.3	57.7	57.7	56.6	55.0	55.4	56.9	53.9	54.7	55.0	54.2
	Japan	48.4	48.7	49.4	50.1	48.9	49.7	48.5	48.5	48.7	50.0	51.5	53.0	51.6	55.1	54.5	54.8
Europe	France	48.5	48.7	49.8	48.1	48.2	50.4	48.2	48.2	47.8	50.7	51.2	50.1	50.0	52.8	49.7	51.2
	Germany	48.3	48.4	48.3	49.0	49.1	49.8	49.5	49.5	48.2	47.0	49.1	50.9	52.2	51.4	50.1	50.3
	Italy	46.6	49.3	49.2	48.4	49.8	50.4	49.0	49.0	50.6	47.9	48.1	50.6	51.3	52.1	52.9	52.2
	Spain	49.5	48.1	50.5	51.4	51.9	54.3	51.5	51.5	51.5	49.6	49.2	50.0	48.7	51.7	51.2	49.7
	United Kingdom	44.9	45.4	46.4	47.7	48.0	47.0	46.2	46.2	50.2	50.6	51.8	51.7	51.0	53.7	53.9	52.5
Americas	Canada	46.3	45.3	46.1	45.6	46.1	48.3	47.7	47.7	48.4	48.6	50.4	51.0	50.0	53.3	52.9	53.0
	Mexico	46.5	44.8	46.7	46.3	49.1	50.2	49.6	49.6	47.3	46.1	46.3	47.1	48.9	47.7	49.6	51.3
	United States	50.2	50.2	52.0	52.9	49.8	53.0	52.0	52.0	52.2	51.8	52.4	51.6	52.3	54.5	55.1	53.9
	US ISM	48.9	48.8	48.6	49.0	48.4	48.9	48.9	48.9	48.0	47.9	52.6	52.4	52.7	52.7	54.0	53.3

Source: Bloomberg Finance L.P.

Table 3: Global light vehicle production growth

Percent change

QoQ	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2024	2025	2026E	2027F
Europe	0.5%	2.4%	-13.4%	12.0%	1.5%	-1.7%	-12.3%	11.6%	0.2%	3.9%	-12.1%	10%				
China	-24.5%	8.0%	7.8%	17.1%	-32.2%	13.2%	8.9%	15.2%	-25.9%	17.1%	-8.2%	26%				
Japan/Korea	-5.4%	-0.4%	-3.3%	6.0%	0.0%	-3.3%	-4.9%	-0.9%	-0.4%	1.4%	-3.5%	6%				
North America	4.2%	5.0%	0.3%	-9.9%	4.0%	4.8%	-0.8%	-9.3%	3.9%	6.9%	-3.3%	-2%				
Other	4.4%	-4.9%	6.9%	1.1%	0.2%	-6.3%	2.0%	-3.0%	3.9%	4.5%	5.1%	-3%				
Global	**-8.9%**	**3.0%**	**0.8%**	**7.4%**	**-11.8%**	**2.9%**	**0.2%**	**5.3%**	**-9.1%**	**8.5%**	**-5.2%**	**10%**				
YoY																
Europe	-5.3%	-0.3%	3.8%	-0.2%	0.8%	-3.3%	-2.0%	-2.4%	-3.6%	1.9%	2.1%	0.3%	-4.4%	-0.7%	-1.7%	0.1%
China	14.5%	11.0%	14.6%	2.8%	-7.6%	-3.1%	-2.1%	-3.6%	5.2%	8.9%	-8.2%	0.4%	3.6%	10.0%	-4.0%	1.2%
Japan/Korea	7.0%	0.3%	0.8%	-3.5%	2.1%	-0.9%	-2.6%	-8.9%	-9.2%	-4.8%	-3.4%	3.0%	-6.4%	1.0%	-2.6%	-3.7%
North America	-5.0%	-3.5%	5.2%	-1.1%	-1.3%	-1.4%	-2.5%	-1.9%	-2.0%	-0.1%	-2.6%	5.8%	-1.5%	-1.2%	-1.8%	0.1%
Other	4.3%	5.6%	1.4%	7.3%	3.0%	1.5%	-3.1%	-7.0%	-3.6%	7.5%	10.7%	11.0%	-0.5%	4.6%	-1.5%	6.2%
Global	**3.9%**	**3.7%**	**6.9%**	**1.6%**	**-1.7%**	**-1.8%**	**-2.4%**	**-4.4%**	**-1.5%**	**3.9%**	**-1.7%**	**3.2%**	**-1.0%**	**3.9%**	**-2.6%**	**1.0%**

*3Q26-4Q27 is a forecast as of June 2026.
Source: S&P Global

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Figure 3: Vehicle sales and penetration rates

Vehicles in thousands

Global PV Sales ('000s)	Jun-26	Jun-25	y/y%	6M26	6M25	y/y%
Passenger Vehicles (PV)						
Europe (EU5)	973	868	12.1%	5,066	4,752	6.6%
Germany	296	256	15.7%	1,484	1,403	5.8%
France	189	170	11.4%	857	842	1.8%
Italy	146	132	10.8%	938	855	9.8%
Spain	128	119	7.8%	648	610	6.2%
United Kingdom	213	191	11.4%	1,138	1,042	9.2%
China	1,602	2,085	-23.2%	8,721	10,895	-20.0%
US	1,379	1,280	7.7%	7,954	8,181	-2.8%
Total	**3,954**	**4,233**	-6.6%	**21,740**	**23,828**	-8.8%
Battery Electric Vehicles (EV)						
Europe (EU5)	233	143	63.1%	1,037	713	45.4%
Germany	84	47	78.2%	368	249	47.9%
France	56	29	91.7%	242	149	62.5%
Italy	15	8	88.3%	80	45	77.8%
Spain	14	11	26.5%	63	46	36.6%
United Kingdom	64	47	35.0%	285	225	26.6%
China	685	661	3.6%	3,106	3,336	-6.9%
US	73	101	-27.6%	458	616	-25.7%
Total	**991**	**905**	9.6%	**4,601**	**4,665**	-1.4%
Plug-in Hybrid Vehicles (PHEV)						
Europe (EU5)	102	82	25.2%	516	394	31.1%
Germany	32	26	25.8%	164	139	17.8%
France	12	12	4.4%	42	49	-15.4%
Italy	16	10	64.6%	85	43	98.6%
Spain	16	14	15.0%	78	56	39.2%
United Kingdom	27	21	24.9%	148	107	38.4%
China	323	450	-28.2%	1,614	2,128	-24.2%
US	18	21	-15.9%	86	175	-51.1%
Total	**443**	**553**	-19.8%	**2,216**	**2,697**	-17.8%

Global EV/PHEV Penetration (%)	Jun-26	Jun-25	y/y%	6M26	6M25	y/y%
EV/PHEV Penetration (%)						
Europe (EU5)	34.5%	25.9%	8.6%p	30.7%	23.3%	7.4%p
Germany	39.2%	28.4%	10.8%p	35.8%	27.6%	8.2%p
France	36.1%	24.1%	12.0%p	33.0%	23.5%	9.5%p
Italy	20.9%	13.2%	7.7%p	17.5%	10.2%	7.3%p
Spain	23.2%	20.8%	2.4%p	21.8%	16.8%	5.0%p
United Kingdom	42.5%	35.9%	6.6%p	38.0%	31.8%	6.2%p
China	62.9%	53.3%	9.6%p	54.1%	50.1%	4.0%p
US	6.6%	9.5%	-2.9%p	6.8%	9.7%	-2.8%p
Total	**36.3%**	**34.4%**	1.8%p	**31.4%**	**30.9%**	0.5%p
EV Penetration (%)						
Europe (EU5)	23.9%	16.4%	7.5%p	20.5%	15.0%	5.5%p
Germany	28.4%	18.4%	10.0%p	24.8%	17.7%	7.1%p
France	29.6%	17.2%	12.4%p	28.2%	17.6%	10.5%p
Italy	10.2%	6.0%	4.2%p	8.5%	5.2%	3.2%p
Spain	11.1%	9.4%	1.6%p	9.8%	7.6%	2.2%p
United Kingdom	30.0%	24.8%	5.2%p	25.0%	21.6%	3.4%p
China	42.8%	31.7%	11.1%p	35.6%	30.6%	5.0%p
US	5.3%	7.9%	-2.6%p	5.8%	7.5%	-1.8%p
Total	**25.1%**	**21.4%**	3.7%p	**21.2%**	**19.6%**	1.6%p
PHEV Penetration (%)						
Europe (EU5)	10.5%	9.4%	1.1%p	10.2%	8.3%	1.9%p
Germany	10.9%	10.0%	0.9%p	11.0%	9.9%	1.1%p
France	6.5%	7.0%	-0.4%p	4.8%	5.8%	-1.0%p
Italy	10.7%	7.2%	3.5%p	9.1%	5.0%	4.0%p
Spain	12.1%	11.4%	0.8%p	12.0%	9.2%	2.9%p
United Kingdom	12.5%	11.2%	1.4%p	13.0%	10.3%	2.7%p
China	20.2%	21.6%	-1.4%p	18.5%	19.5%	-1.0%p
US	1.3%	1.6%	-0.4%p	1.1%	2.1%	-1.1%p
Total	**11.2%**	**13.1%**	-1.9%p	**10.2%**	**11.3%**	-1.1%p

*As of May 2026.
Source: CNEV Post, European Union, US Motor Intelligence, J.P. Morgan. Note: China data are based on preliminary CPCA retail sales data.

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China demand indicators

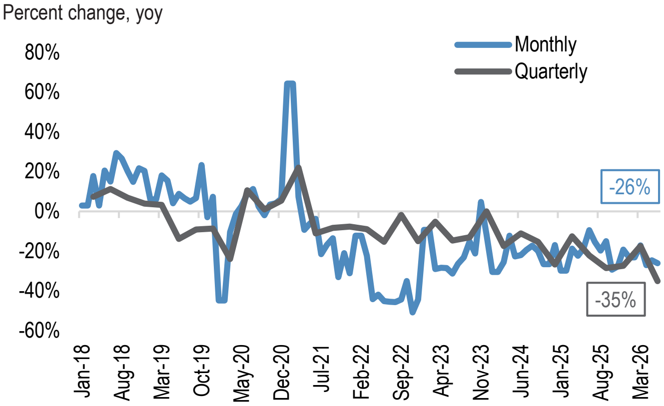
Table 4: China metals demand indicator heat map

	China End-Use Demand Indicators Heatmap	Units	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26
Property	70 cities newly built price index	YoY	-4.6%	-4.1%	-3.7%	-3.4%	-3.0%	-2.7%	-2.6%	-2.8%	-3.1%	-3.3%	-3.5%	-3.6%	-3.7%	-3.6%	-3.5%
	Residential building starts (million sqm)	YTD YoY	-22.3%	-21.4%	-19.6%	-18.3%	-18.3%	-18.3%	-19.3%	-19.9%	-19.8%	-23.3%	-23.3%	-22.0%	-23.6%	-23.4%	-24.1%
	Residential area under const. (million sqm)	YTD YoY	-10.1%	-9.6%	-9.4%	-9.4%	-9.6%	-9.7%	-9.7%	-10.0%	-10.3%	-11.9%	-11.9%	-12.1%	-12.5%	-12.6%	-12.9%
	Residential const. completed (m. sqm)	YTD YoY	-16.8%	-17.6%	-15.5%	-17.3%	-18.5%	-17.1%	-18.9%	-20.1%	-20.2%	-26.9%	-26.9%	-26.5%	-25.8%	-25.0%	-25.3%
Infrastructure	FAI - total (RMB)	YTD YoY	4.0%	3.7%	2.8%	1.6%	0.5%	-0.5%	-1.7%	-2.6%	-3.8%	1.8%	1.8%	1.7%	-1.6%	-4.1%	-5.7%
	FAI- manufacturing (RMB)	YTD YoY	8.8%	8.5%	7.5%	6.2%	5.1%	4.0%	2.7%	1.9%	0.6%	3.1%	3.1%	4.1%	1.2%	-0.4%	-1.2%
	FAI- electric equipment and machinery (RMB)	YTD YoY	-7.5%	-8.6%	-7.8%	-8.7%	-8.8%	-9.5%	-9.4%	-9.5%	-10.3%	5.8%	5.8%	0.8%	2.9%	1.5%	1.2%
	FAI- railway transportation (RMB)	YTD YoY	1.6%	2.3%	4.2%	5.9%	4.5%	4.2%	3.0%	2.7%	-1.2%	28.7%	28.7%	12.6%	8.6%	5.1%	0.2%
	FAI- construction (RMB)	YTD YoY	-0.3%	-19.8%	-21.2%	-24.7%	-24.7%	-17.2%	-13.7%	-14.7%	-22.2%	2.2%	2.2%	0.1%	-11.7%	-6.0%	-14.8%
	FAI- real estate (RMB)	YTD YoY	-10.6%	-11.1%	-11.5%	-12.4%	-13.2%	-14.0%	-14.9%	-16.0%	-17.5%	-10.7%	-10.7%	-10.6%	-13.4%	-15.9%	-18.0%
	FAI- investment in grid infrastructure (RMB)	YTD YoY	14.6%	19.8%	14.6%	12.5%	14.0%	9.9%	7.2%	5.9%	5.1%	92.1%	92.1%	43.4%	32.4%	13.2%	4.0%
	Cement production (million tonnes)	YTD YoY	-5.3%	-8.1%	-5.3%	-5.6%	-6.2%	-8.6%	-15.8%	-8.2%	-6.6%	6.8%	6.8%	-21.0%	-10.8%	-8.1%	-5.6%
Manufacturing	China NBS Manufacturing PMI	Index	49.0	49.5	49.7	49.3	49.4	49.8	49.0	49.2	50.1	49.3	49.0	50.4	50.3	50.0	50.3
	Refrigerator Production (units)	YTD YoY	-2.0%	-2.6%	0.3%	1.7%	3.5%	3.8%	3.1%	3.6%	5.1%	10.1%	10.1%	12.7%	15.9%	16.5%	14.1%
	Washing Machine Production (units)	YTD YoY	10.5%	9.3%	10.3%	10.9%	9.5%	9.2%	8.2%	8.1%	6.6%	0.3%	0.3%	3.6%	3.9%	3.5%	2.8%
	Airconditioner Production (units)	YTD YoY	6.2%	4.7%	4.0%	3.9%	4.3%	3.0%	2.5%	1.2%	0.4%	-2.8%	-2.8%	0.2%	-2.4%	-3.6%	-4.7%
	Freezer Production (units)	YTD YoY	-7.0%	-5.7%	-2.1%	0.5%	2.3%	1.9%	-0.7%	0.0%	0.8%	20.5%	20.5%	14.5%	16.2%	18.4%	16.4%
	Total Automotive Production (units)	YoY	8.9%	11.6%	11.4%	13.3%	13.0%	17.1%	12.1%	2.8%	-2.1%	0.01%	-20.5%	-3.0%	-1.7%	-1.2%	-1.2%
	Automotive Sales for Passenger Vehicles (units)	YoY	11.0%	13.3%	14.5%	14.7%	16.5%	13.2%	7.5%	1.2%	-8.7%	-6.8%	-15.4%	-2.3%	-4.2%	-4.2%	-5.3%
	Automotive Sales for Commercial Vehicles (units)	YoY	2.7%	-2.0%	9.5%	14.1%	16.3%	29.6%	21.0%	24.4%	15.3%	23.5%	-14.0%	8.9%	8.1%	12.5%	10.7%
	Automotive Sales for Heavy-Duty Truck (units)	YoY	37.3%	43.3%	42.3%	83.5%	53.5%	71.5%	55.4%	60.6%	47.9%	47.7%	14.5%	23.3%	16.3%	17.6%	21.1%
	Automotive Production for Heavy-Duty Truck (units)	YoY	31%	40%	40%	76%	64%	64%	80%	96%	69%	70%	20%	23%	23%	13%	6%
	Sales of BEV Passenger Vehicles	YoY	60.5%	45.9%	39.5%	39.7%	37.4%	33.1%	29.8%	23.6%	14.0%	0.4%	-17.6%	-4.2%	-0.8%	24.3%	30.7%

Source: Bloomberg Finance L.P., NBS, J.P. Morgan Commodities Research

Figure 4: Building new starts growth

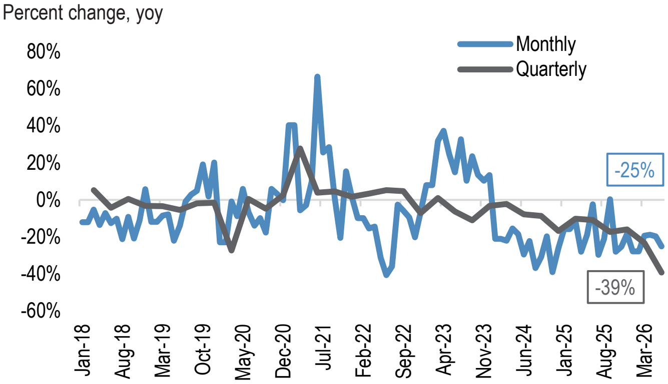
Percent change, yoy



*Includes residential and non-residential. Text box highlights the latest data point; monthly is through June 2026 and quarterly is as of 2Q26.
Source: China NBS

Figure 5: Building completions growth

Percent change, yoy



*Includes residential and non-residential. Text box highlights the latest data point; monthly is through June 2026 and quarterly is as of 2Q26.
Source: China NBS

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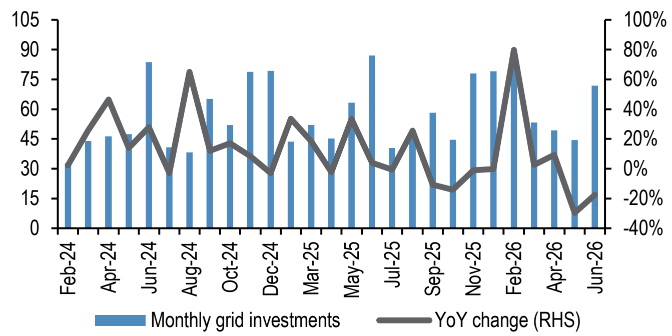
Figure 6: Fixed asset investment in grid infrastructure

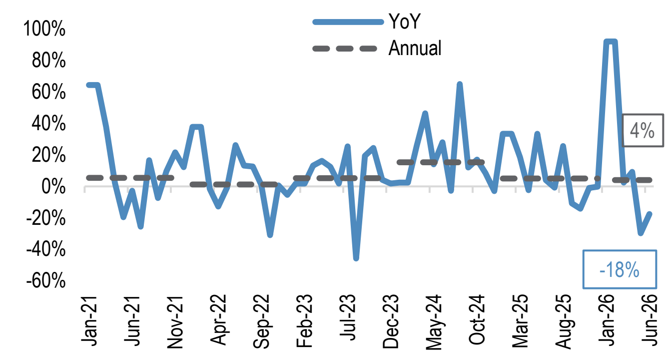
LHS: CNY billions; RHS: Percent change, yoy



*Monthly data through 2025 from NEA. 2026 onwards from Wind. Feb = Jan & Feb combined.
Source: China National Energy Administration, Wind

Figure 7: Fixed asset investment in grid infrastructure

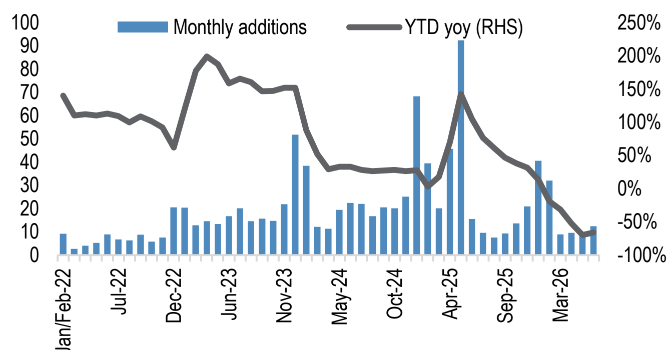
Percent change, yoy



*Monthly data through 2025 from NEA. 2026 onwards from Wind.
Source: China National Energy Administration, Wind.

Figure 8: Solar power plant capacity additions

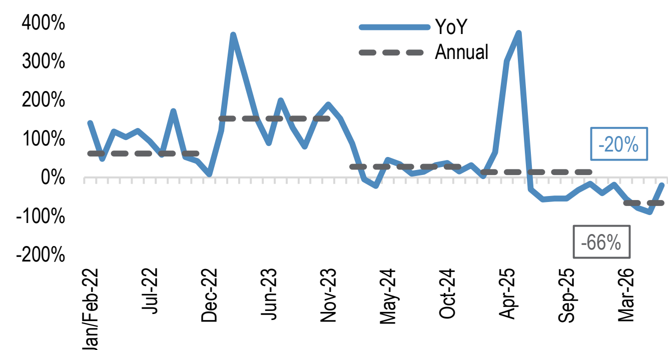
LHS: Gigawatts; RHS: Percent change, YTD yoy



*2026 is through June.
Source: China National Energy Administration

Figure 9: Solar power plant capacity addition growth

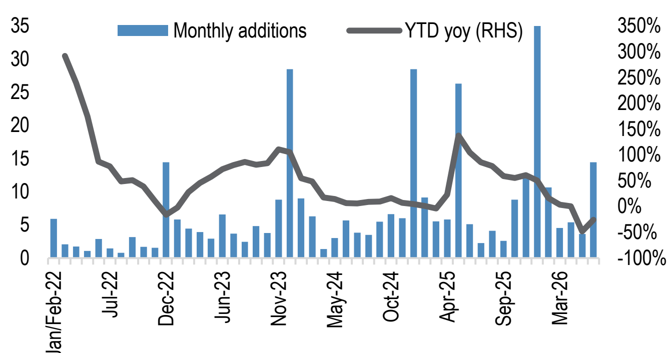
Percent change, yoy



*2026 is through June.
Source: China National Energy Administration

Figure 10: Wind power plant capacity additions

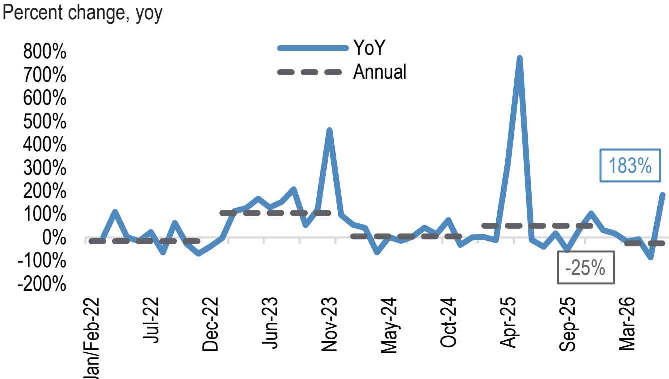
LHS: Gigawatts; RHS: Percent change, yoy



*2026 is through June.
Source: China National Energy Administration

Figure 11: Wind power plant capacity addition growth

Percent change, yoy



*2026 is through June.
Source: China National Energy Administration

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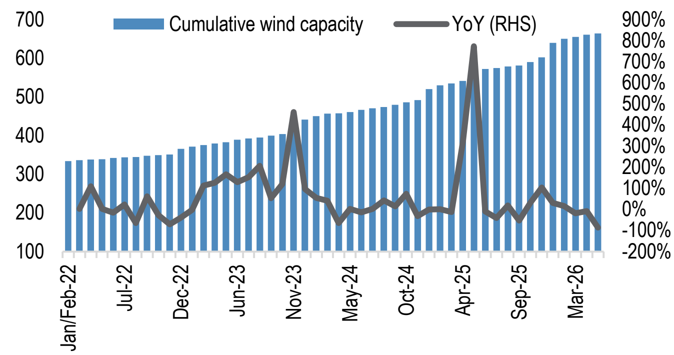
Figure 12: Cumulative wind installed capacity

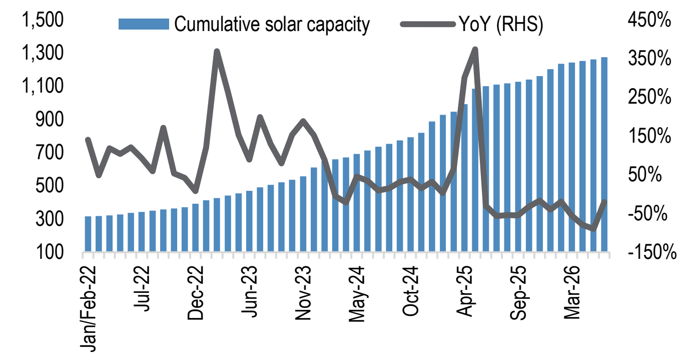
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through June.
Source: China National Energy Administration

Figure 13: Cumulative solar installed capacity

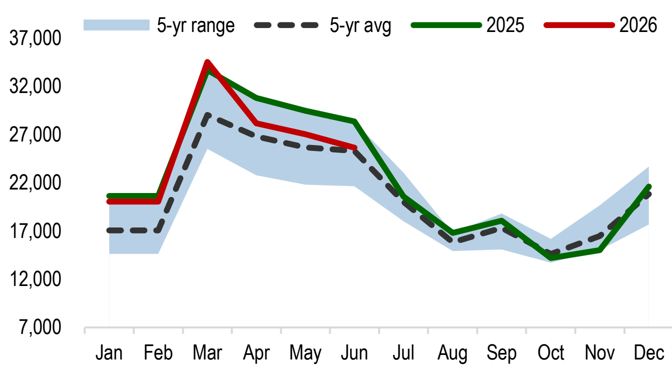
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through June.
Source: China National Energy Administration

Figure 14: Air conditioner production, monthly

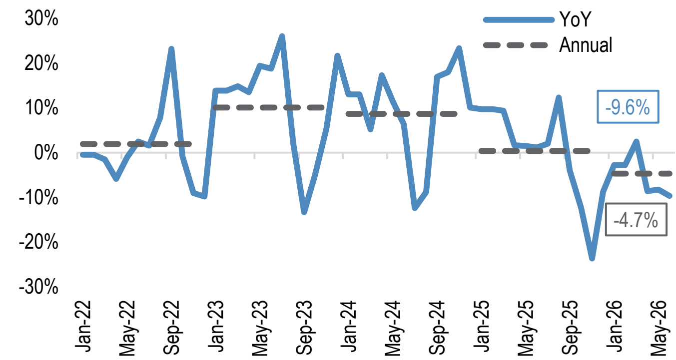
Thousand units



Source: China NBS

Figure 15: Air conditioner production growth

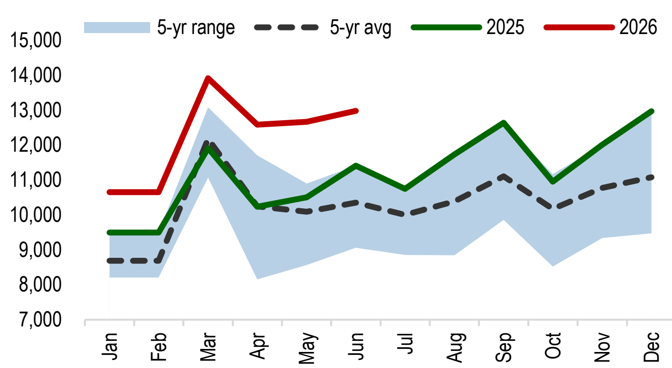
Percent change, yoy



*2026 is through June.
Source: China NBS

Figure 16: Fridge and freezer production, monthly

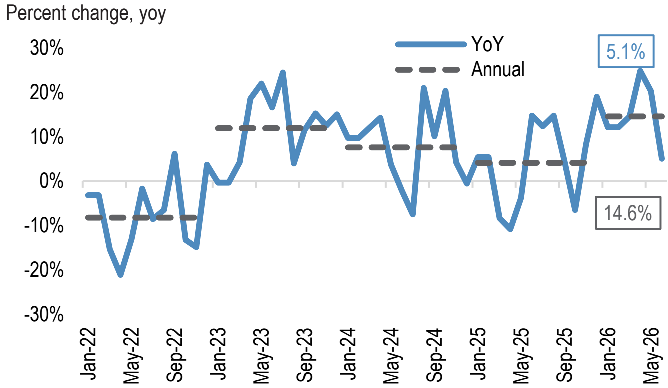
Thousand units



Source: China NBS

Figure 17: Fridge and freezer production growth

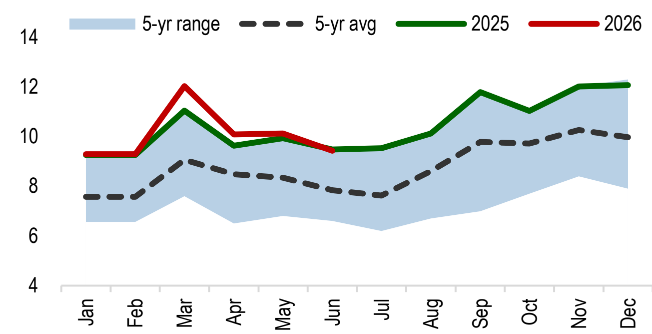
Percent change, yoy



*2026 is through June.
Source: China NBS

Figure 18: Washing machine production, monthly

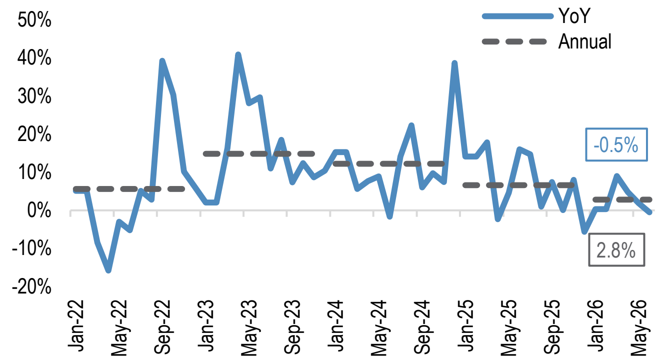
Million units



Source: Source: China NBS

Figure 19: Washing machine production growth

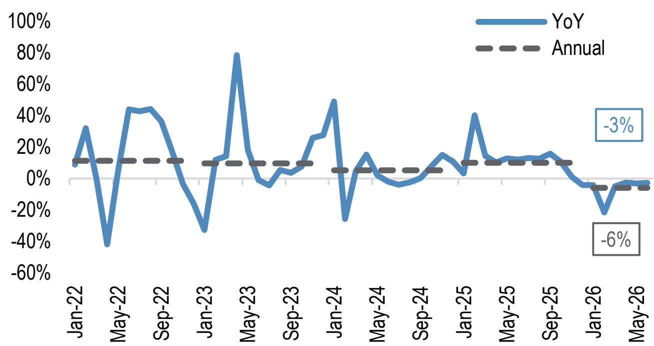
Percent change, yoy



*2026 is through June.
Source: China NBS

Figure 20: Passenger vehicle production growth

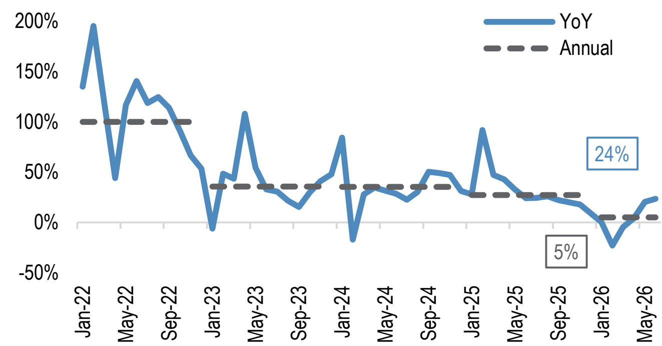
Percent change, yoy



*Includes basic passenger, multi-purpose, SUVs, and crossover vehicles. 2026 is through June.
Source: China Automotive and Research Center

Figure 21: Passenger NEV production growth

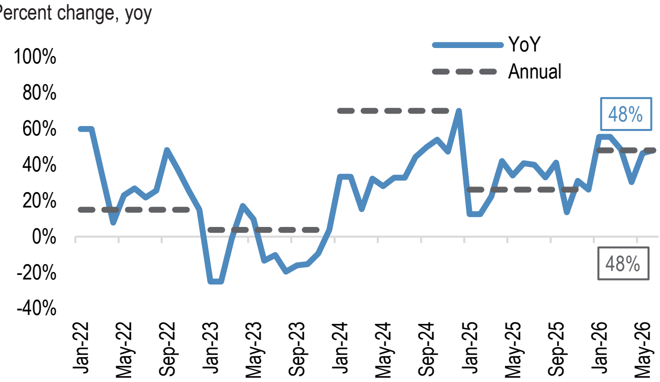
Percent change, yoy



*Includes battery electric vehicles, and plug-in hybrids. 2026 is through June.
Source: China Automotive and Research Center

Figure 22: Industrial robot production growth

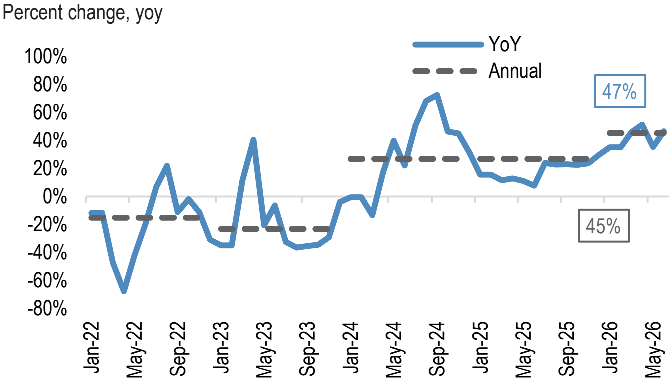
Percent change, yoy



*2026 is through June.
Source: China NBS

Figure 23: Excavator production growth

Percent change, yoy



*2026 is through June.
Source: China NBS

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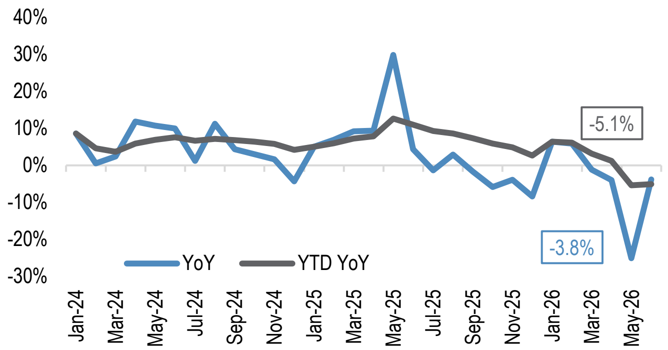
Copper

Figure 24: Consumption-weighted end use indicator for Chinese copper demand growth

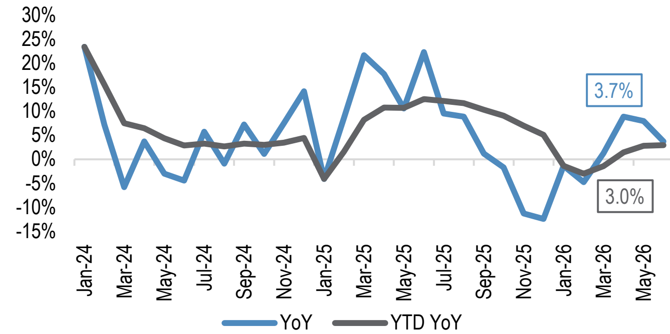
LHS: Percent change, yoy; RHS: Percent change, YTD yoy



*This indicator applies consumption-weights to underlying reported end-use production/installation/export growth figures to derive an estimate for monthly aggregate Chinese copper demand growth. It covers construction, grid, solar, wind, EVs, other transport, consumer durables, ACs/refrigeration, machinery and product and cable exports which, combined, roughly account for 90% of Chinese end use demand. Data through June 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 25: China copper apparent demand growth

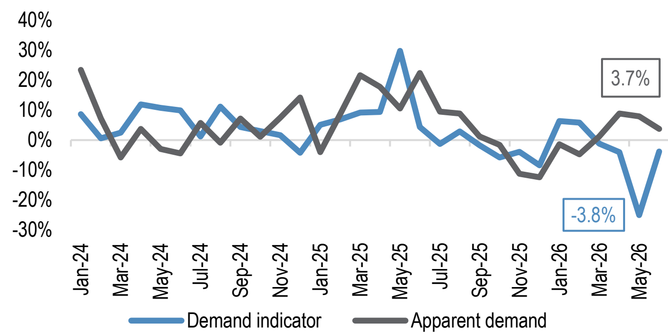
Percent change



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE. Data through June 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 26: YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

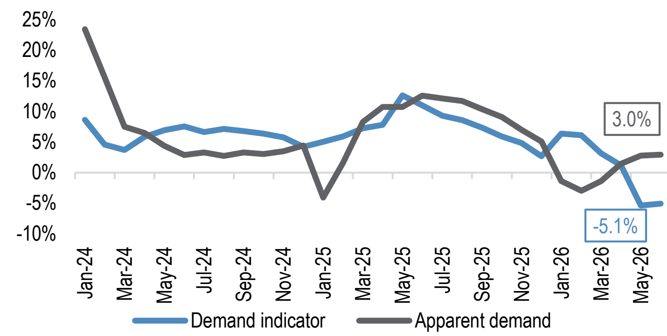
Percent change, yoy



*See explanatory notes in Figures 22 & 23. Data through June 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 27: YTD YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

Percent change, YTD yoy



*See explanatory notes in Figures 22 & 23. Data through June 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

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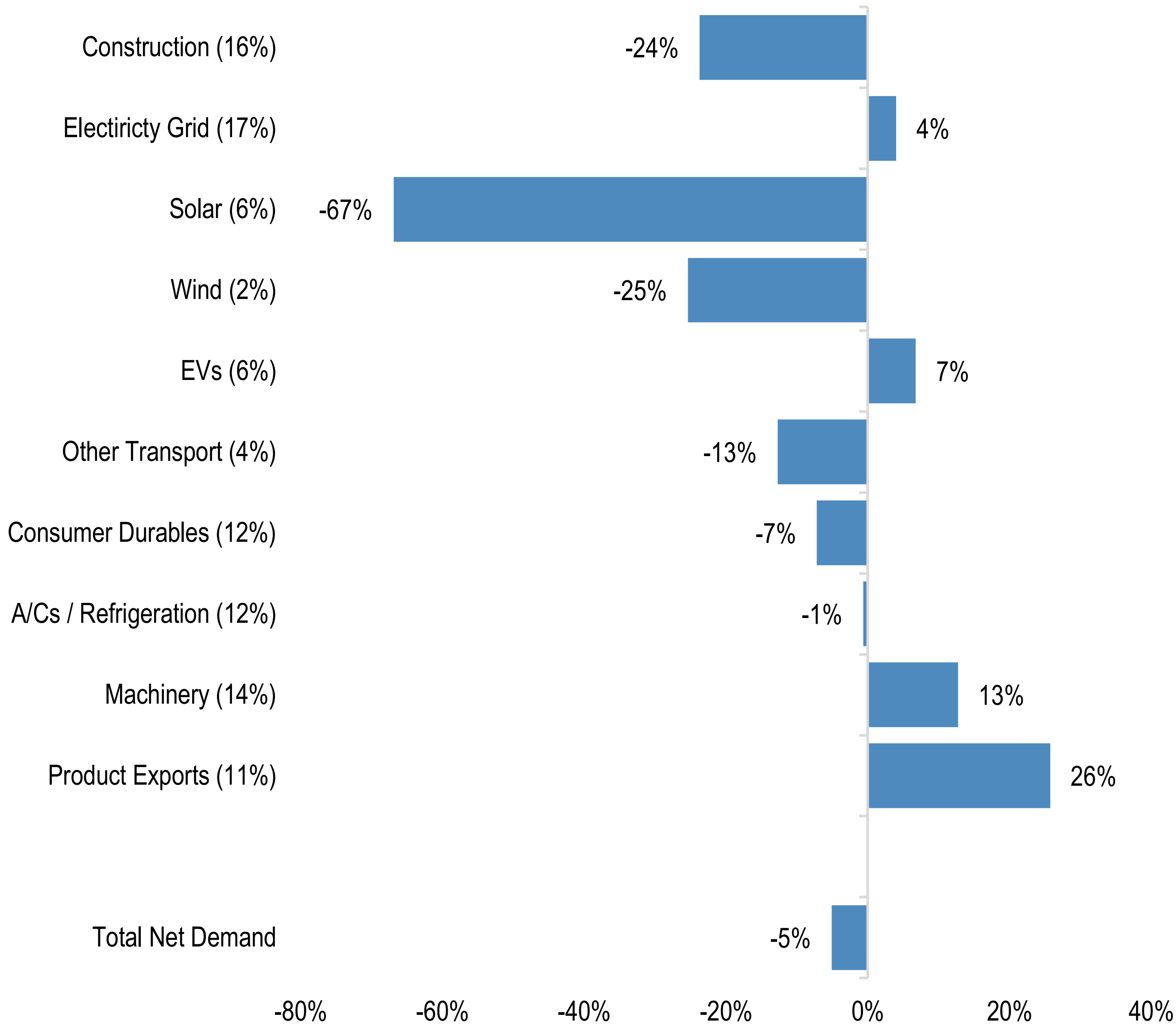
Figure 28: Consumption-weighted end use indicator for YTD Chinese copper demand growth by sector

Percent change, YTD yoy (2025 end-use weights in parenthesis).

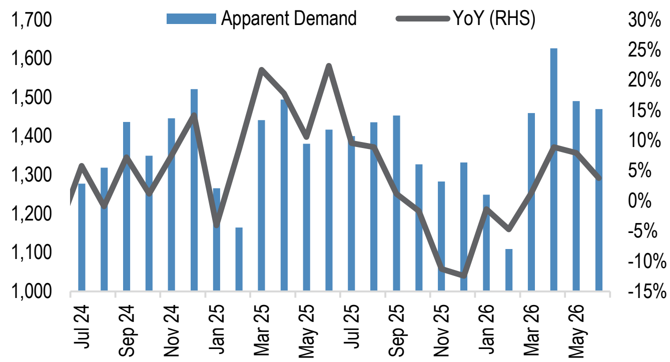


*See explanatory notes in Figures 22 & 23. Data through June 2026.

Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 29: China apparent copper demand

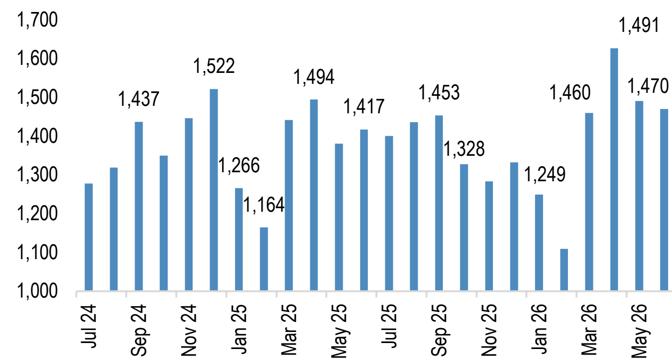
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 30: China apparent copper demand

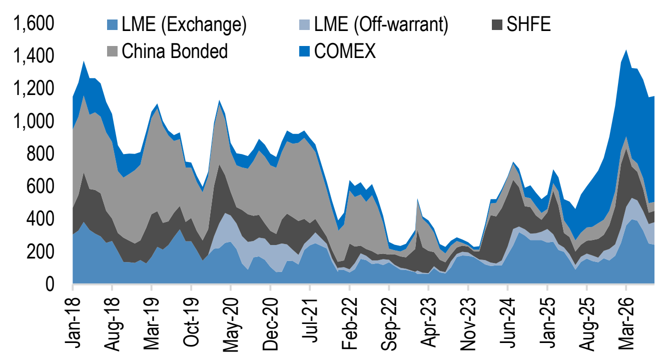
Thousand mt



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 31: Global visible copper inventories

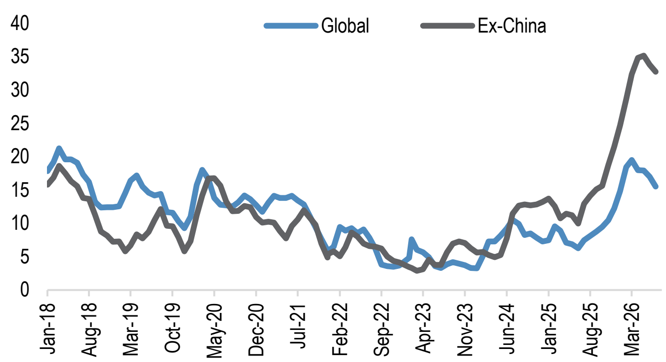
Thousand mt



Source: CRU, LME, SHFE, COMEX. Bonded includes Chinese bonded stocks in Shanghai, Guangdong and Qingdao. As of 3rd August 2026, while LME off-warrant as of 29th July 2026.

Figure 32: Global visible copper inventory in days of use

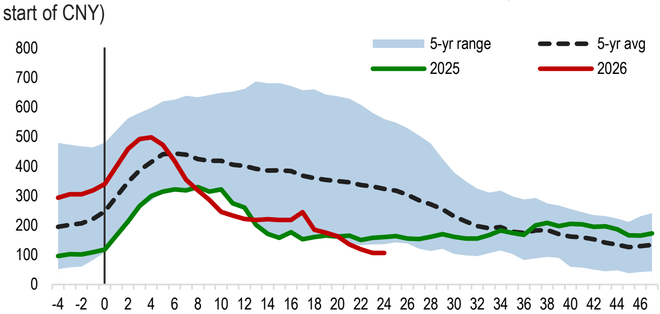
Days



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 33: Total visible China copper inventory (SHFE + Bonded)

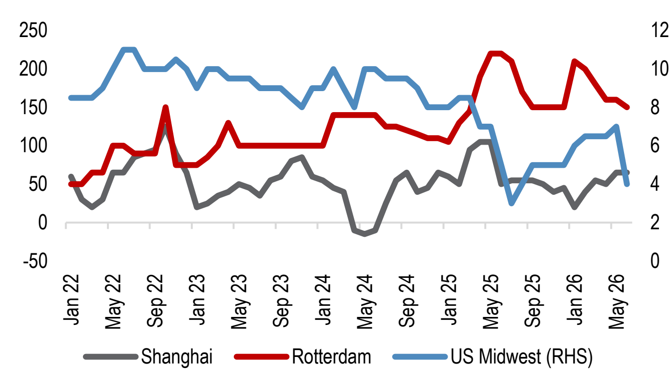
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 34: Copper regional premiums

LHS: US$/t; RHS: US$c/lb



Source: CRU

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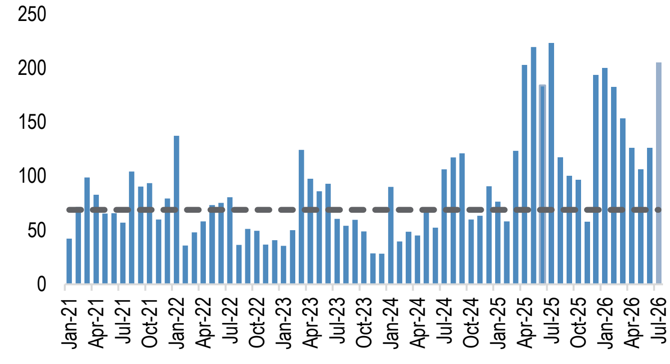
Figure 35: Monthly US refined copper imports

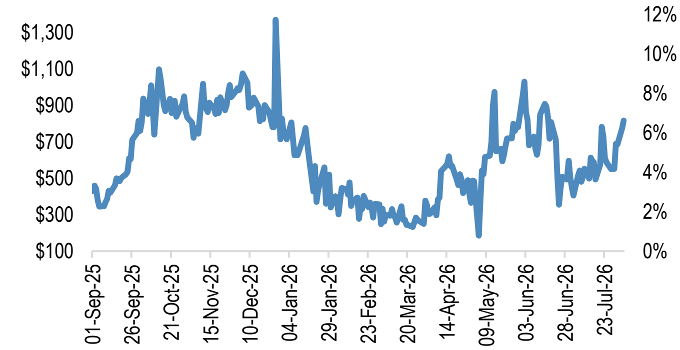
LHS: Thousand mt, RHS: US$/mt



Note: July 2026 estimated based on bill of lading data.
Source: US Census Bureau, IHS Markit, COMEX, LME, J.P. Morgan Commodities Research

Figure 36: Dec'26 COMEX/LME copper spread

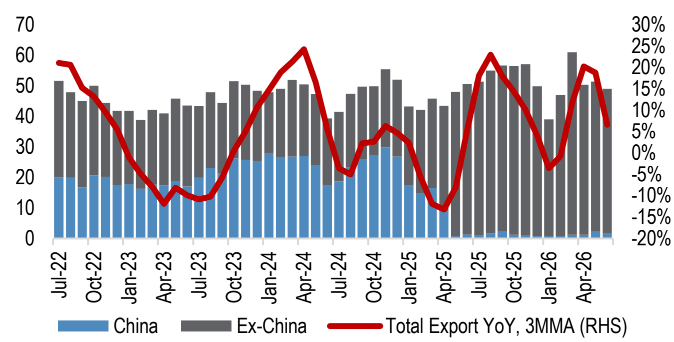
LHS: US$/mt, RHS: Percent of LME price (implied tariff rate)



Source: Bloomberg Finance L.P. *Arb as of London, 4th August 2026.

Figure 37: Monthly US copper scrap exports

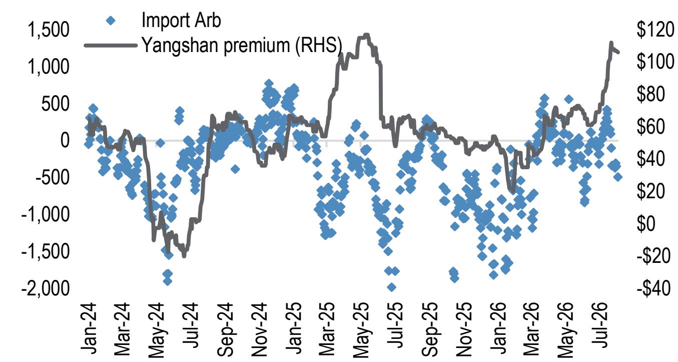
LHS: Thousand mt (cu contained); RHS: Percent change, YoY



*Data as of June 2026.
Source: US Census Bureau, J.P. Morgan Commodities Research

Figure 38: Chinese spot copper import arbitrage and onshore spot premium/discount

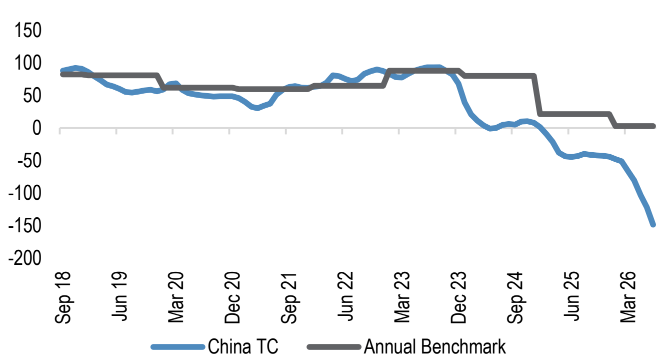
LHS: CNY/mt ; RHS: US$/mt



Source: SMM, Bloomberg Finance L.P., J.P. Morgan Commodities Research. *Arb as of 8:00 AM London. As of 3rd August 2026.

Figure 39: China copper imported concentrate treatment charges vs annual benchmark

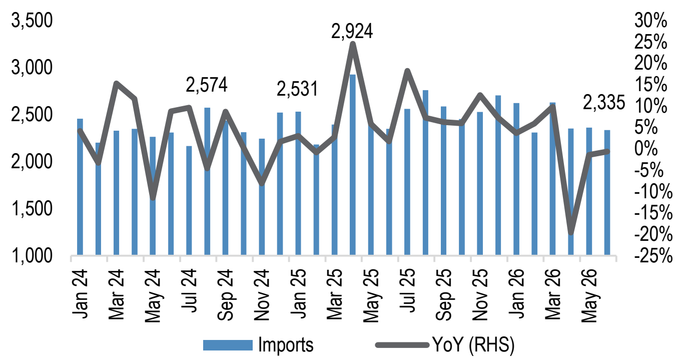
US$/mt



Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 40: China copper concentrate imports

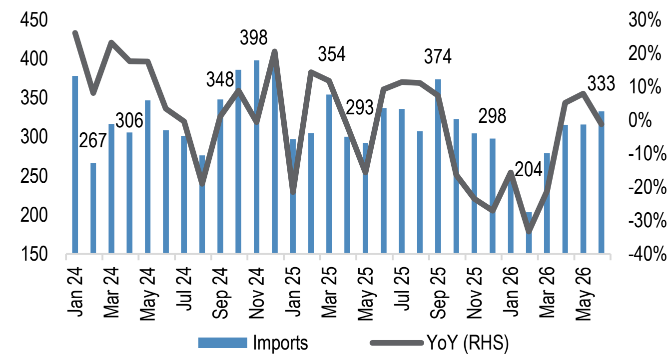
LHS: Thousand mt (gross); RHS: Percent change, yoy



Source: China Customs General Administration

Figure 41: China refined copper imports

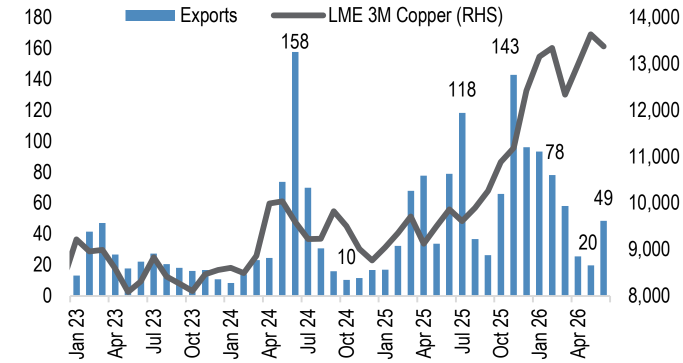
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration

Figure 42: China refined copper exports vs LME 3-month copper price

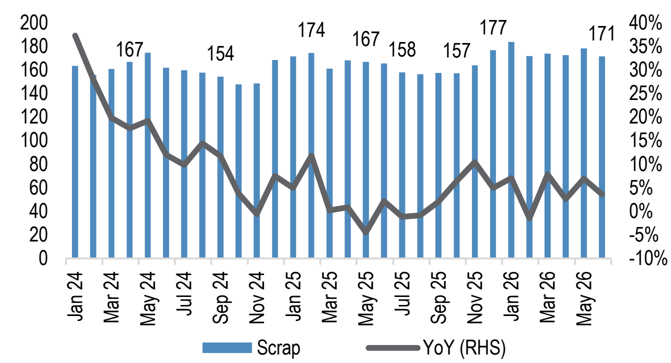
LHS: Thousand mt; RHS: US$/mt



Source: China Customs General Administration, Bloomberg Finance L.P.

Figure 43: China contained copper scrap imports, 3mma

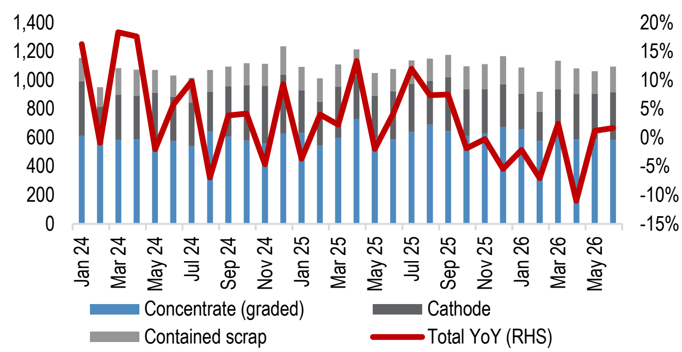
LHS: Thousand mt (Cu contained); RHS: Percent change, yoy



Source: China Customs General Administration

Figure 44: China copper imports of concentrate, cathode and contained scrap

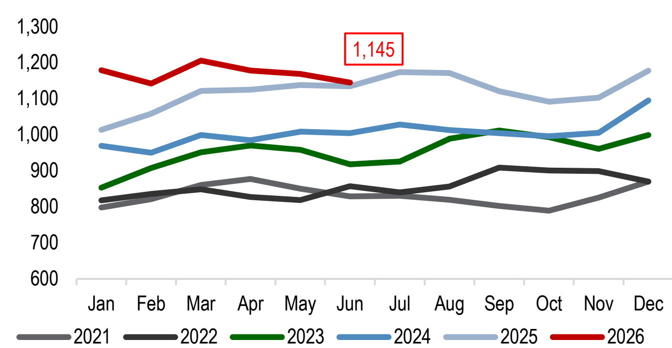
LHS: Thousand mt; RHS: Percent change, yoy



*Assumes 25% copper contained in concentrate.
Source: China Customs General Administration

Figure 45: China refined copper production, monthly

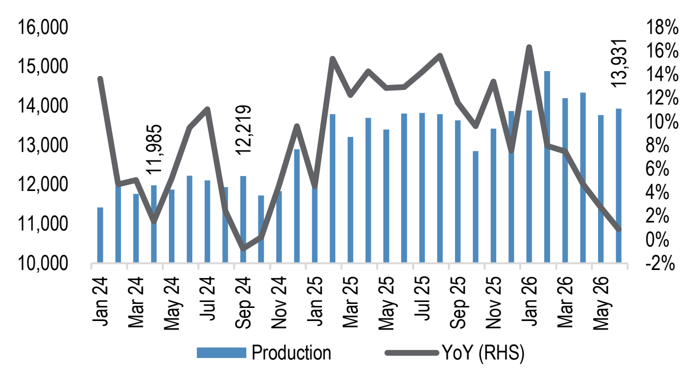
Thousand mt



Source: SMM

Figure 46: Annualized China refined copper production

LHS: Thousand mt; RHS: Percent change, yoy



Source: SMM

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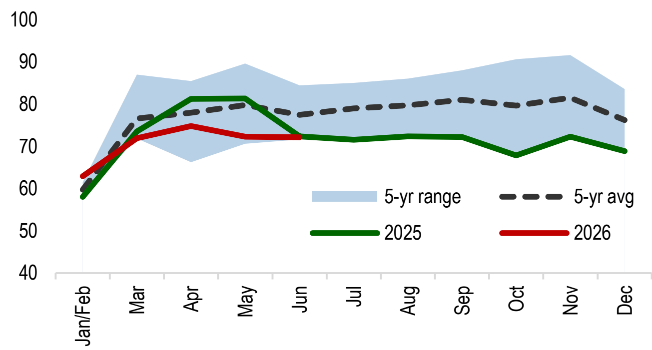
Figure 47: China copper wire and cable producer operating rates

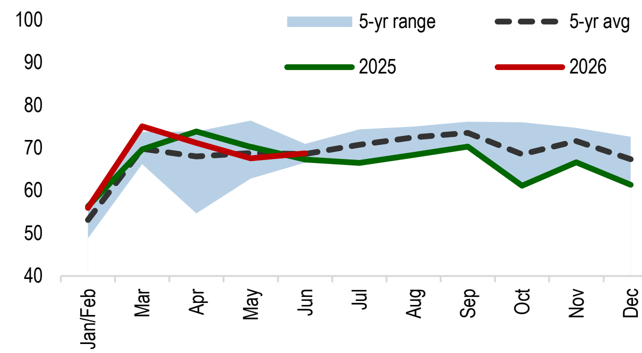
Percent



Source: SMM

Figure 48: China copper cathode-fed rod producer operating rates

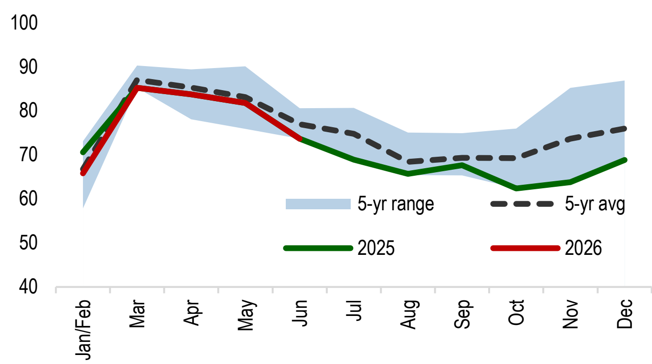
Percent



Source: SMM

Figure 49: China copper tube producer operating rates

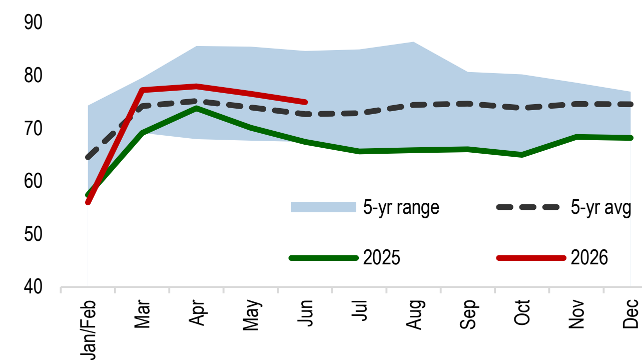
Percent



Source: SMM

Figure 50: China copper flat-rolled producer operating rates

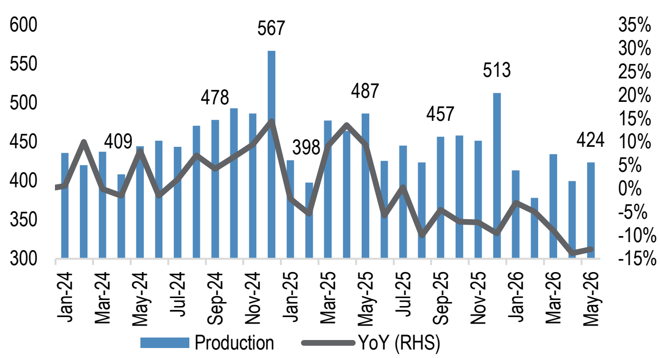
Percent



Source: SMM

Figure 51: Chile copper mine production

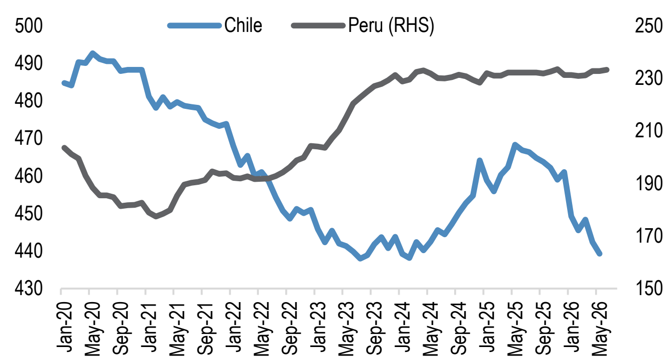
LHS: Thousand metric tonnes; RHS: Percent change yoy



Source: Instituto National de Estadistico

Figure 52: Chile and Peru copper mine production, 12mma

Thousand metric tonnes



Source: Instituto National de Estadistico, Ministerio de Energias y Minas

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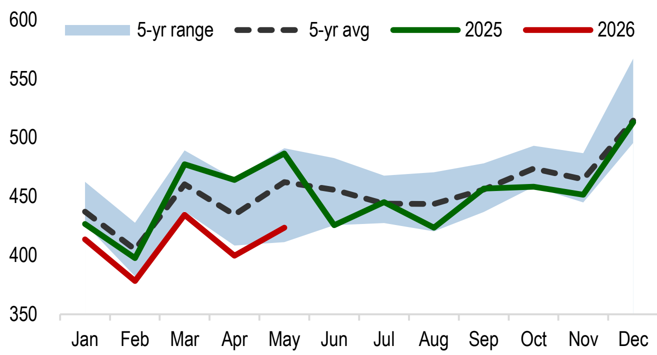
Figure 53: Chile copper mine production, monthly

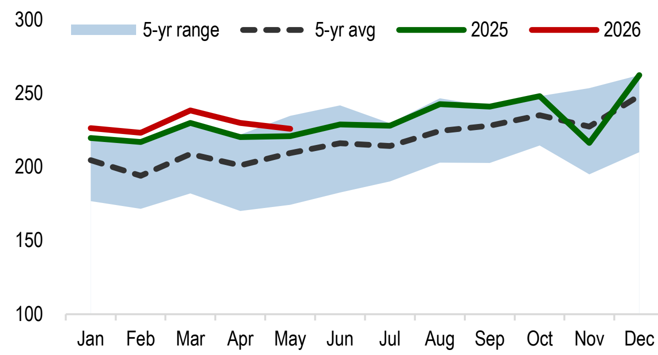
Thousand mt



Source: Instituto National de Estadistico

Figure 54: Peru copper mine production, monthly

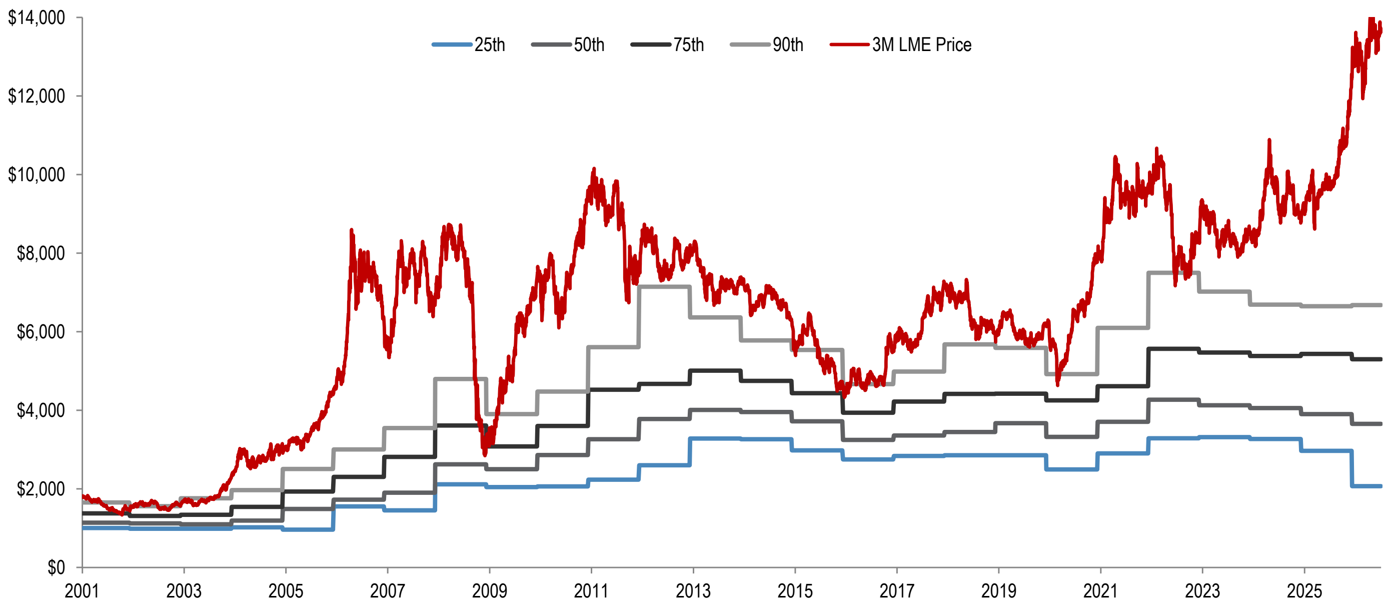
Thousand mt



Source: Ministerio de Energias y Minas

Figure 55: Global copper mining C1 + sustaining capex costs by %-tile and LME 3M copper price

US$/mt MOD



Source: Wood Mackenzie, J.P. Morgan Commodities Research. *LME prices as of 29th July 2026.

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Table 5: China copper trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Copper Concentrate Imports (Gross)	19,720	22,024	21,777	23,428	25,318	27,588	28,165	30,365	14,611
Refined Copper Imports	3,753	3,550	4,670	3,627	3,885	3,739	4,042	3,830	1,699
Unwrought Copper and Products Imports	5,293	4,991	6,686	5,530	5,872	5,504	5,700	5,350	2,500
Contained Copper Imports (from Scrap)	1,434	1,239	733	1,360	1,588	1,721	1,937	1,983	1,036
Refined Copper Exports	282	316	190	266	253	279	458	795	324
Unwrought Copper & Copper Products Exports	793	841	676	933	917	958	1,287	1,673	879
Net Refined Imports	3,472	3,234	4,480	3,361	3,632	3,459	3,584	3,035	1,375
YoY-%									
Copper Concentrate Imports		11.7%	-1.1%	7.6%	8.1%	9.0%	2.1%	7.8%	-1.1%
Refined Copper Imports		-5.4%	31.5%	-22.3%	7.1%	-3.8%	8.1%	-5.3%	-10.0%
Unwrought Copper and Products Imports		-5.7%	34.0%	-17.3%	6.2%	-6.3%	3.6%	-6.1%	-5.3%
Contained Copper Imports (from scrap)		-13.6%	-40.8%	85.4%	16.8%	8.4%	12.5%	2.4%	5.7%
Refined Copper Exports		12.4%	-40.1%	40.5%	-5.2%	10.6%	63.9%	73.6%	5.1%
Unwrought Copper & Copper Products Exports		6.1%	-19.6%	38.0%	-1.8%	4.5%	34.3%	30.0%	17.3%
Net Refined Imports		-6.8%	38.5%	-25.0%	8.1%	-4.8%	3.6%	-15.3%	-12.9%

*Assumes 25% copper contained in concentrate. 2026 is through June.
Source: China Customs General Administration

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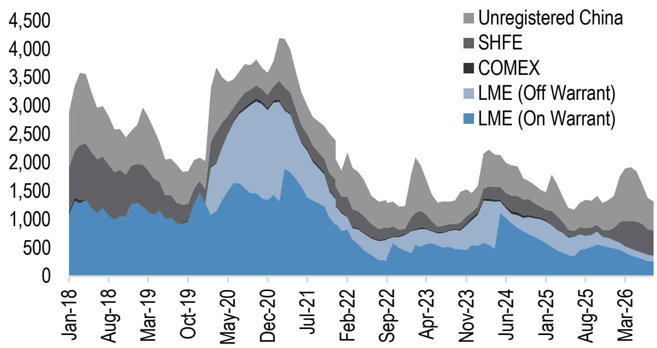
Aluminium

Figure 56: Global visible aluminium inventories

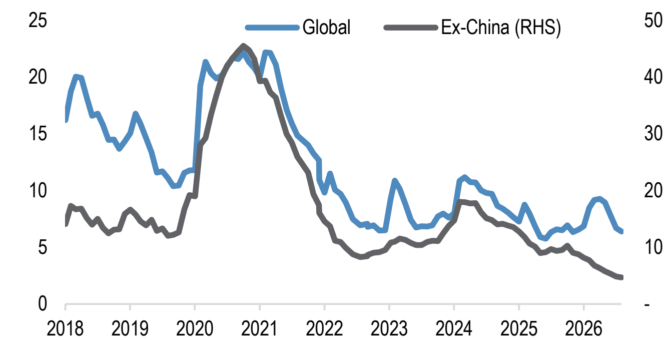
Thousand mt



Source: CRU, LME, SHFE, COMEX. Regional includes stocks in Shanghai, Jiangsu, Nanhai, Zhejiang, Tianjin, Henan, Shandong, Chongqing. As of 3rd August 2026.

Figure 57: Global visible aluminium inventory in days of use

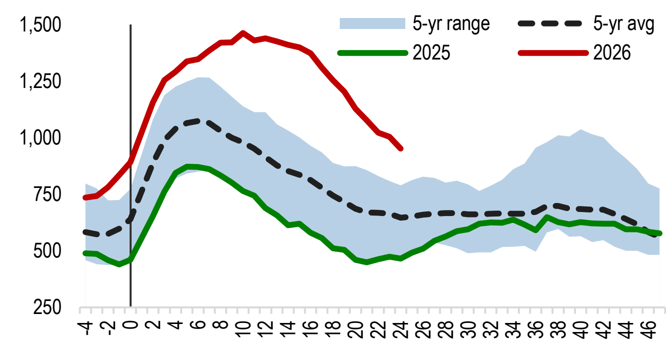
Days



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 58: Total visible China aluminium inventory (SHFE + regional warehouses)

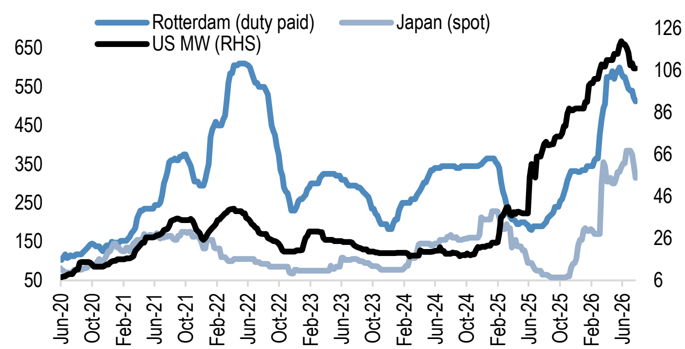
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 59: Regional aluminium ingot premiums

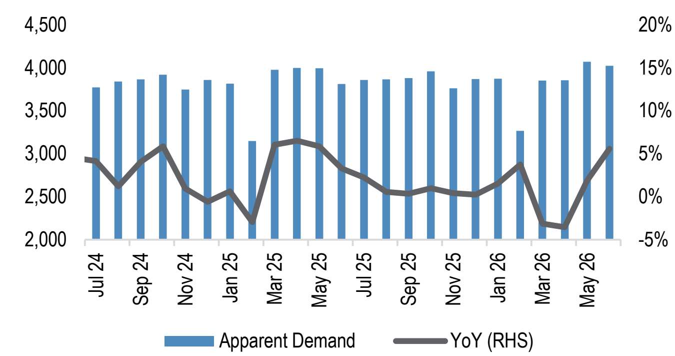
LHS: US$/mt; RHS: c/lb



Source: CRU

Figure 60: China apparent aluminium demand

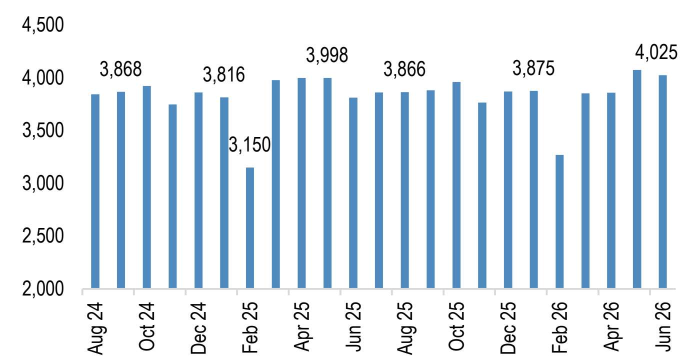
LHS: Thousand mt; RHS: Percent change, yoy



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.
Source: CRU, China Customs General Administration, SHFE

Figure 61: China apparent aluminium demand

Thousand metric tonnes



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.
Source: CRU, China Customs General Administration, SHFE

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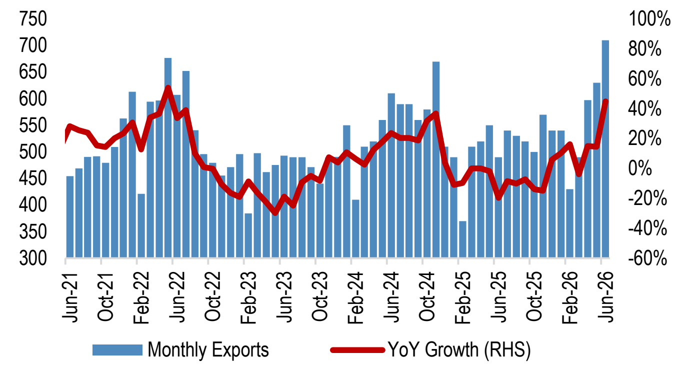
Figure 62: Chinese unwrought aluminium and aluminium product exports

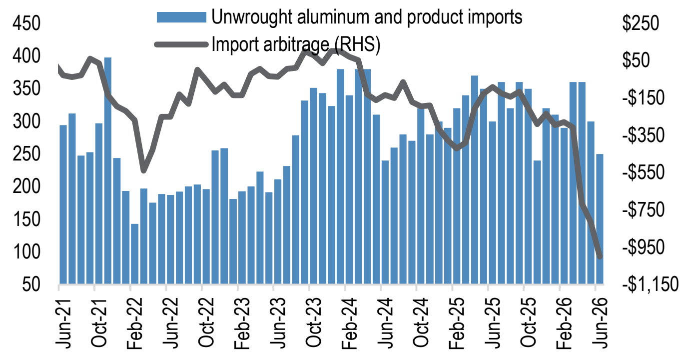
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs, J.P. Morgan Commodities Research

Figure 63: Chinese unwrought aluminium and aluminium product imports

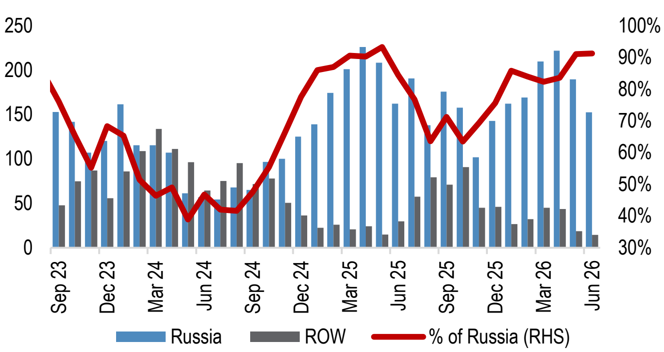
LHS: Thousand mt; RHS: US$/mt



Source: China Customs, J.P. Morgan Commodities Research

Figure 64: China refined aluminium imports by region

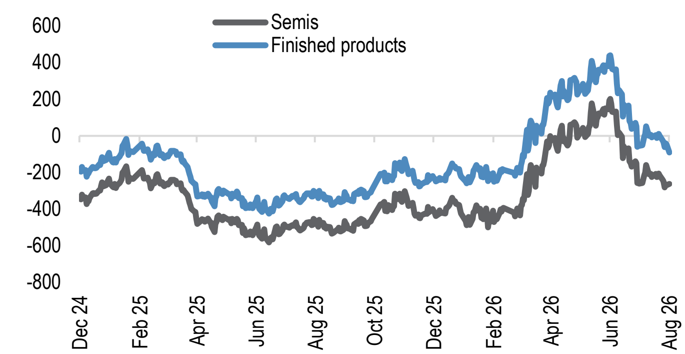
LHS: Thousand mt; RHS: Percent of total



Source: China Customs, J.P. Morgan Commodities Research

Figure 65: China aluminium semis and finished products export arb proxy (China to rest of Asia)

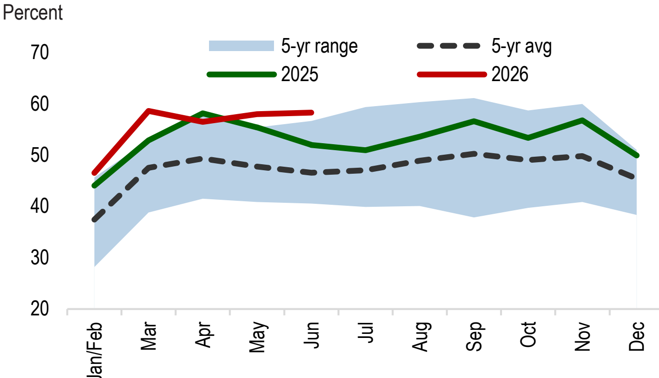
US$/mt. Differential between China spot prices (inclusive of VAT) and LME + Asia premium. Finished product = wire.



Note: Export arb is calculated as the differential between China spot prices and LME + Asia premium less conversion costs (ingot to semis/wire and semis to finished products), logistics, shipping and financing costs.
Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 4th August 2026.

Figure 66: China aluminium wire & rod producer operating rates

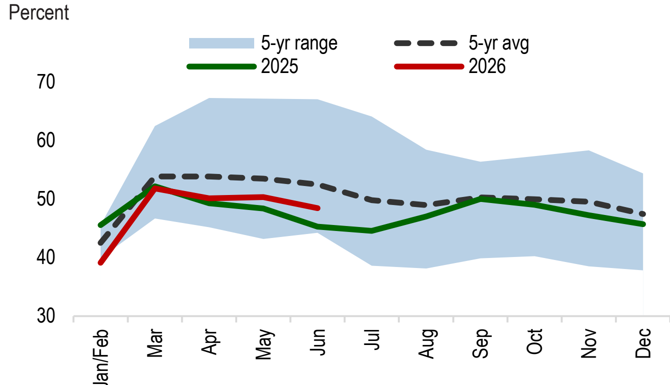
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 67: China aluminium profile producer operating rates

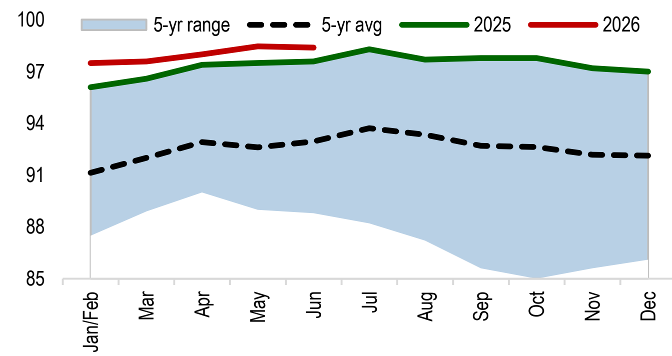
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 68: China aluminium ingot producer operating rates

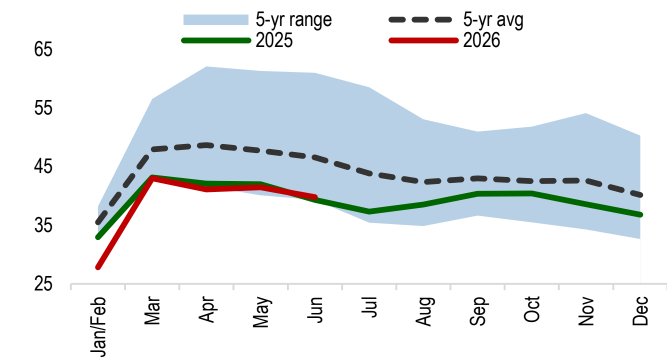
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 69: China aluminium extrusions producer operating rates

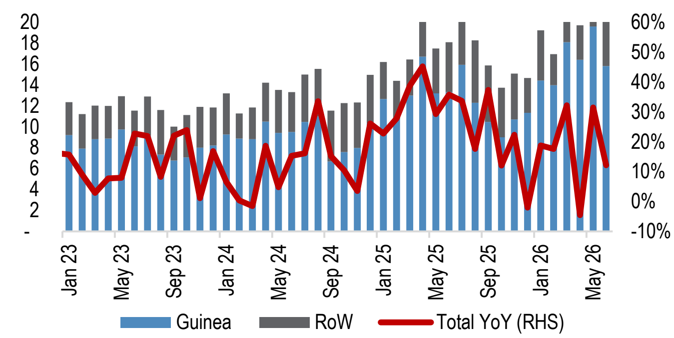
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 70: China total bauxite imports

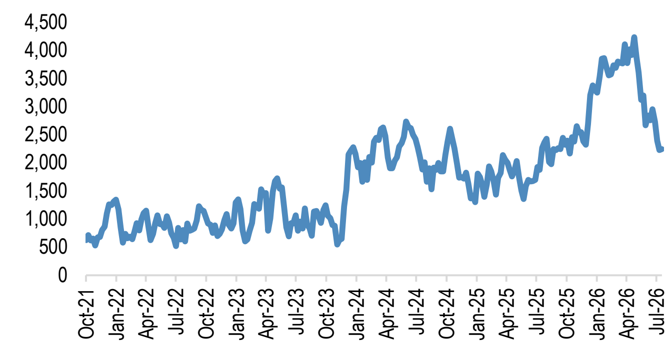
LHS: Million mt; RHS: Percent change yoy



Source: China Customs

Figure 71: China imports of bauxite from Guinea, 3-week ma

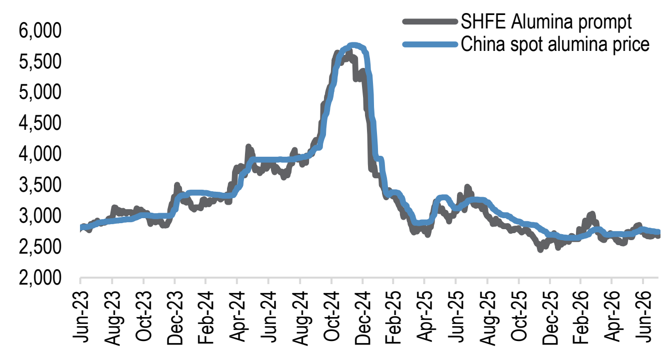
Thousand mt



Source: Mysteel Global

Figure 72: China alumina price

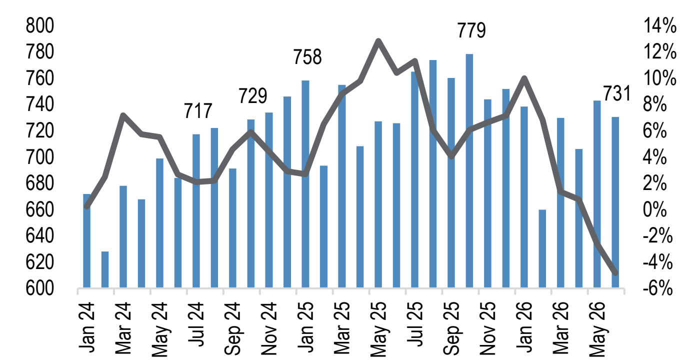
CNY/mt



Source: SHFE, Asian Metal Inc, Bloomberg Finance L.P.

Figure 73: China monthly alumina production

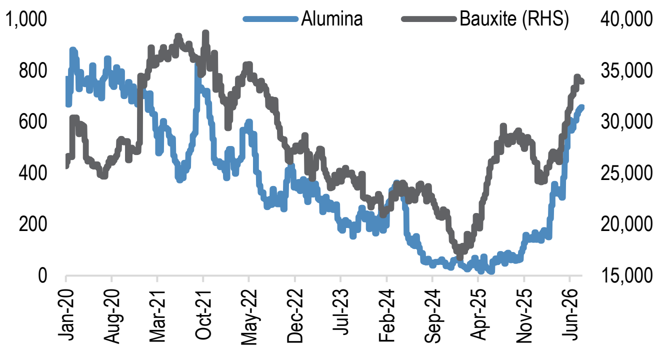
LHS: Thousand mt; RHS: Percent change



Source: SMM, Bloomberg Finance L.P.

Figure 74: China port inventory of bauxite and alumina

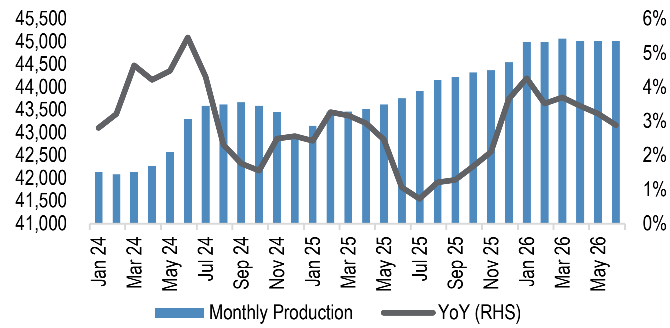
Thousand mt



Source: Mysteel Global

Figure 75: Annualized Chinese refined monthly aluminium production

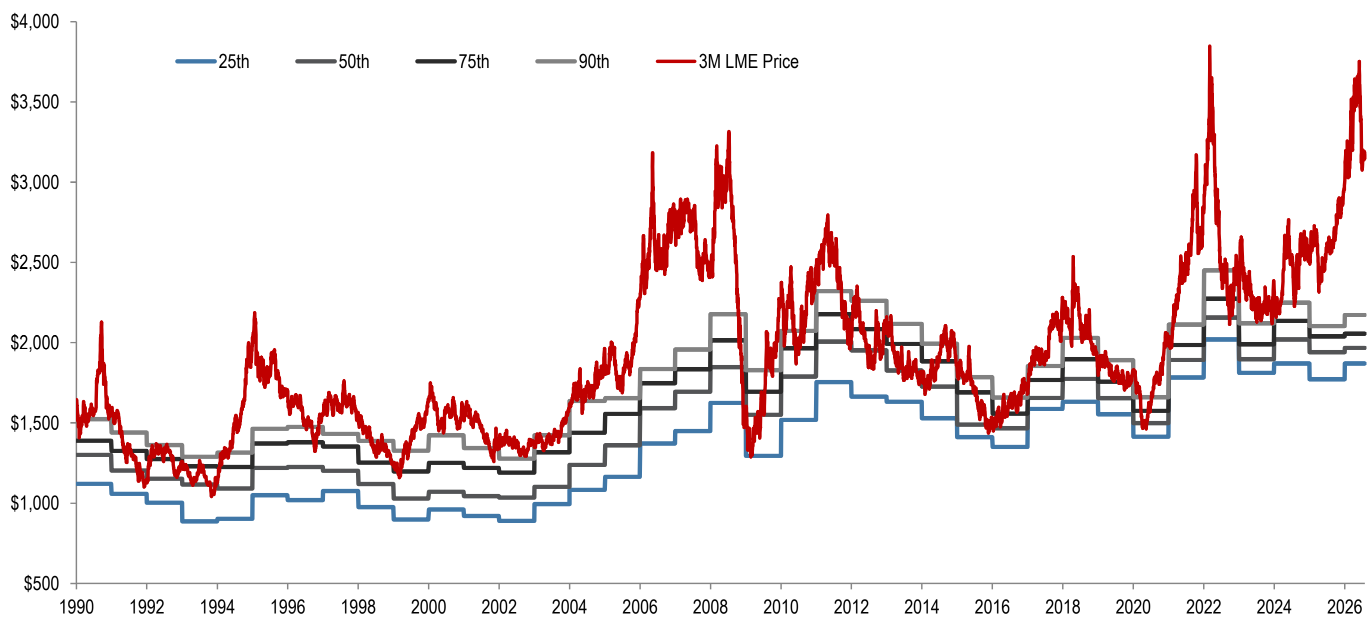
LHS: Thousand mt; RHS: Percent change



Source: CRU

Figure 76: Global primary aluminium smelting C1 costs by %-tile and LME 3M aluminium price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 29th July 2026.

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Table 6: China aluminium trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Bauxite Imports	82,689	100,665	111,537	107,421	125,689	141,647	159,179	201,185	121,079
Refined Aluminium Imports	125	75	1,064	1,580	668	1,543	2,136	2,547	1,286
Unwrought Aluminium and Products Imports	596	646	2,704	3,214	2,391	3,060	3,740	3,920	1,870
Aluminum Waste and Scrap Imports	1,565	1,395	825	1,033	1,518	1,753	1,785	2,016	133
Refined Aluminium Exports	55	76	10	7	196	150	121	298	38
Unwrought Aluminium & Aluminium Products Exports	5,800	5,735	4,860	5,620	6,604	5,679	6,660	6,130	3,398
Net Refined Imports	70	-1	1,055	1,573	472	1,393	2,015	2,249	1,248
YoY-%									
Bauxite Imports		21.7%	10.8%	-3.7%	17.0%	12.7%	12.4%	26.4%	17.1%
Refined Aluminium Imports		-39.5%	1309.8%	48.4%	-57.7%	131.0%	38.4%	19.2%	2.9%
Unwrought Aluminium and Products Imports		8.3%	318.8%	18.8%	-25.6%	28.0%	22.2%	4.8%	-5.1%
Aluminum Waste and Scrap Imports		-10.8%	-40.9%	25.3%	46.9%	15.5%	1.8%	12.9%	-86.9%
Refined Aluminium Exports		38.0%	-87.3%	-27.1%	2680.0%	-23.4%	-19.4%	145.7%	-96.2%
Unwrought Aluminium & Aluminium Products Exports		-1.1%	-15.3%	15.6%	17.5%	-14.0%	17.3%	-8.0%	16.0%

*2026 is through June.
Source: China Customs General Administration

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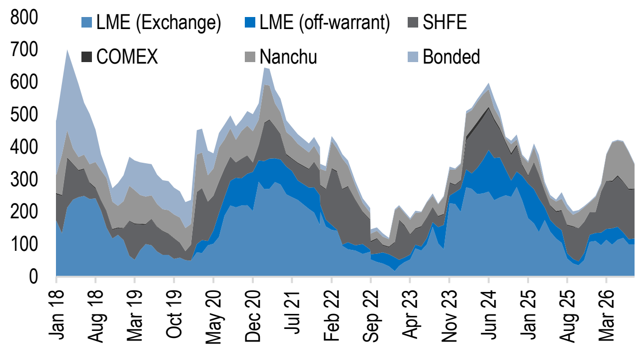
Zinc

Figure 77: Global visible zinc inventories

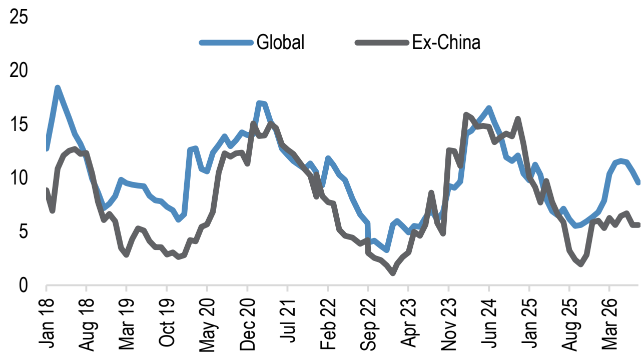
Thousand metric tonnes



Source: CRU, LME, SHFE, COMEX. Nanchu and Bonded warehouses are in China. As of 3rd August 2026.

Figure 78: Global visible zinc inventories in days of use

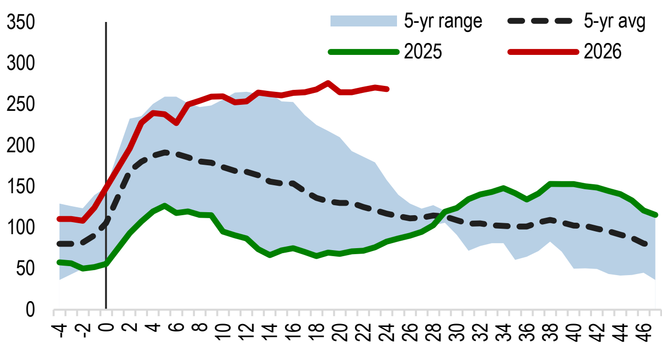
Days



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 79: Total visible China zinc inventory (SHFE + Regional Warehouses)

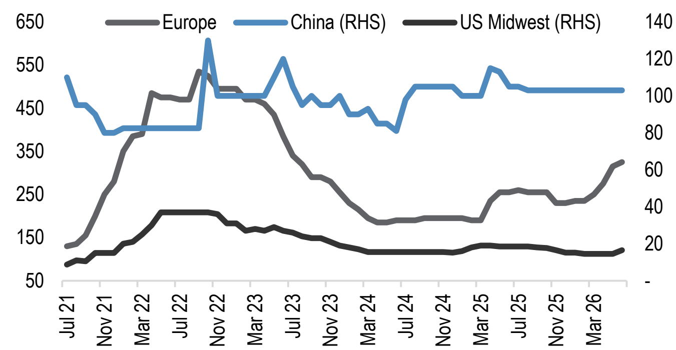
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 80: Regional zinc premiums

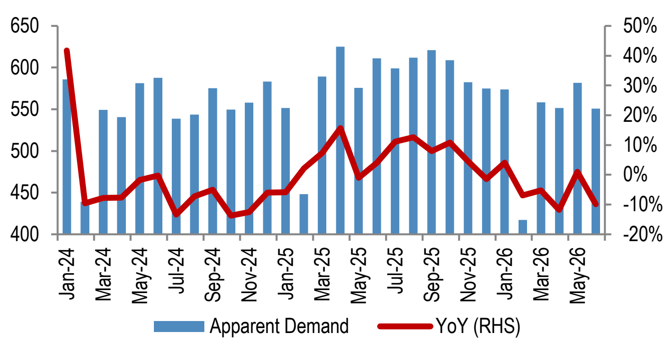
US$/mt



Source: CRU

Figure 81: China apparent zinc demand

LHS: Thousand mt; RHS: Percent change, yoy

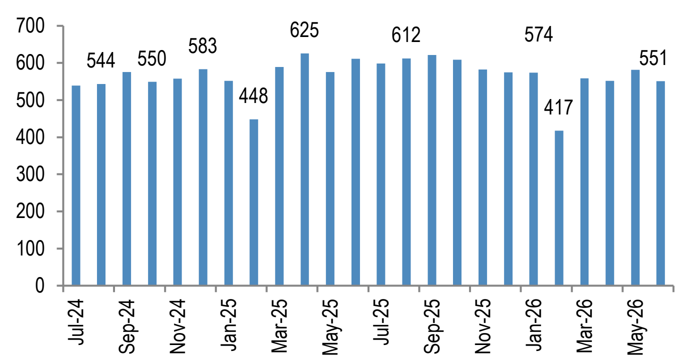


*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE

Source: China Customs General Administration, SHFE

Figure 82: China apparent zinc demand

Thousand mt



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Source: China Customs General Administration, SHFE

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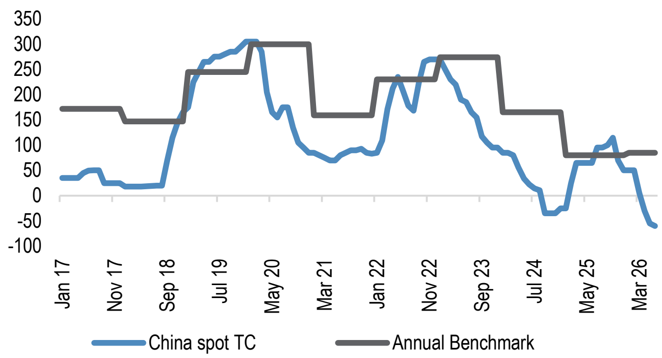
Figure 83: China zinc spot concentrate treatment charges vs annual benchmark

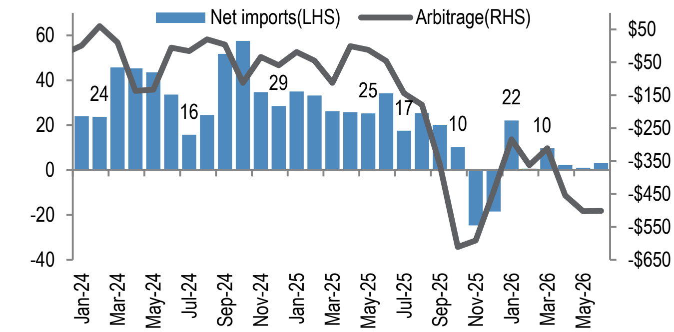
US$/mt



Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 84: China net imports of refined zinc and China's zinc import arb

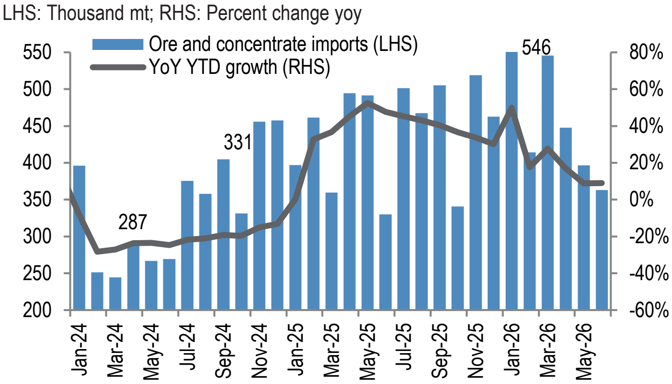
LHS: Thousand mt; RHS: US$/mt



Source: China Customs, CRU, J.P. Morgan Commodities Research

Figure 85: China zinc concentrate imports

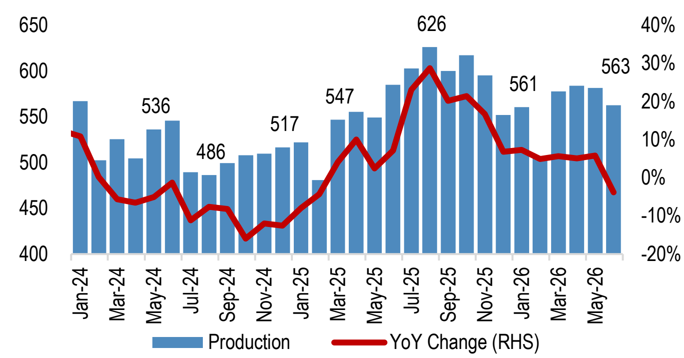
LHS: Thousand mt; RHS: Percent change yoy



Source: NBS, Bloomberg Finance L.P.

Figure 86: China monthly refined zinc production

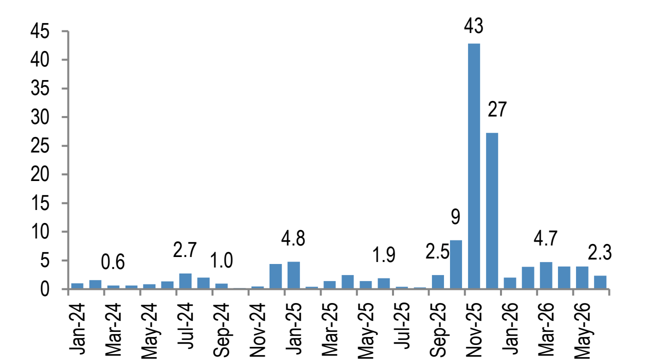
LHS: Thousand mt; RHS: Percent change yoy



Source: SMM, Bloomberg Finance L.P.

Figure 87: China refined zinc exports

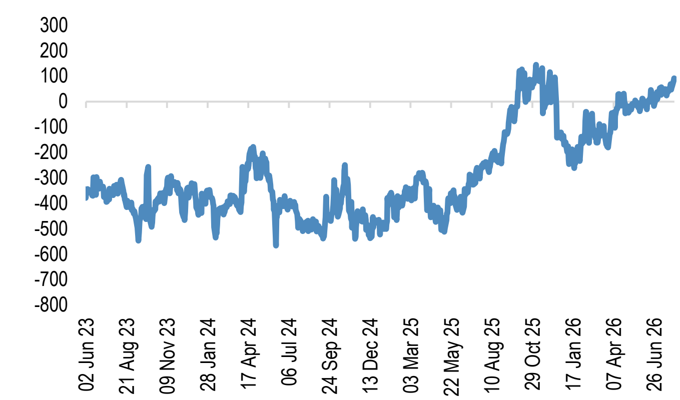
Thousand mt



Source: NBS, Bloomberg Finance L.P.

Figure 88: China zinc export arb proxy

US$/mt. Differential between China spot prices (inclusive of VAT) and LME prompt futures + Asia premium



Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 4th August 2026.

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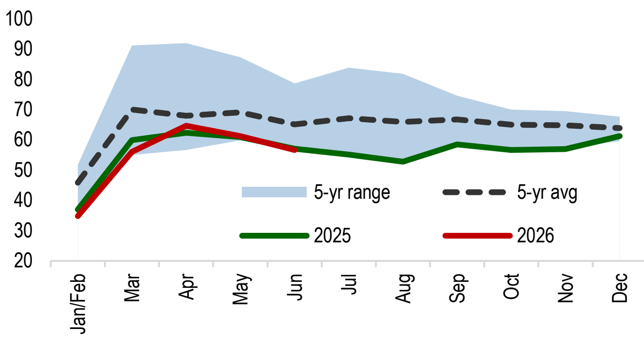
Figure 89: China zinc galvanizers operating rates

Percent

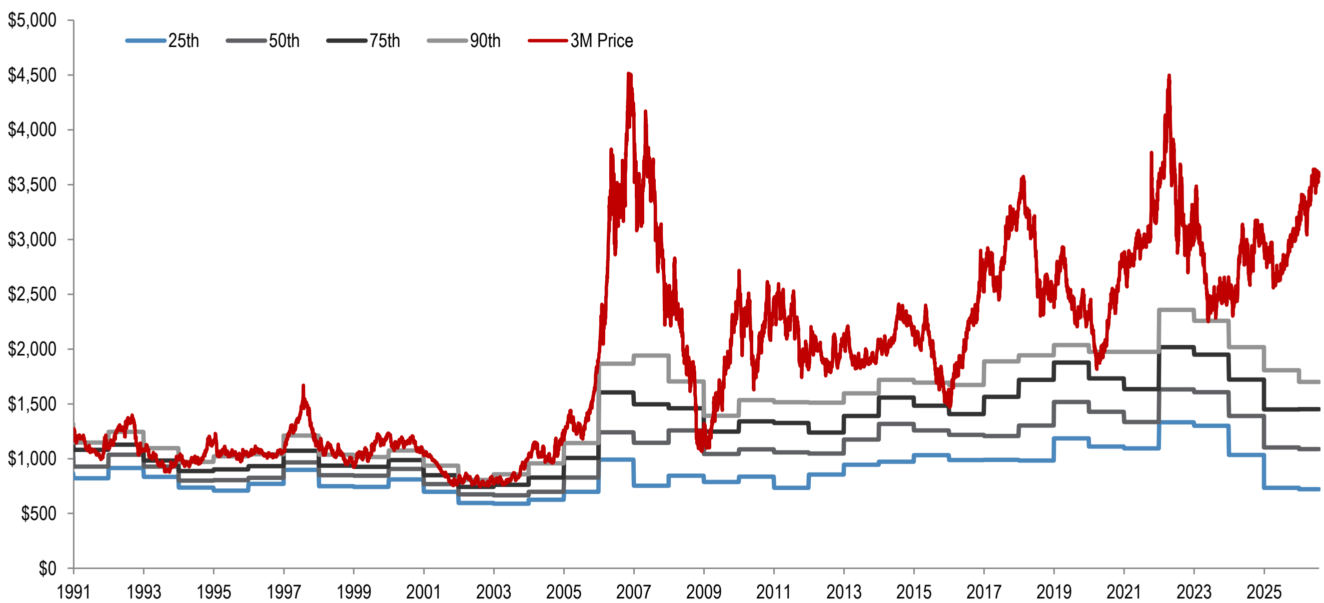


Source: SMM, J.P. Morgan Commodities Research

Source: .

Figure 90: Global zinc C1 mining costs by %-tile and LME 3M copper price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 29th July 2026.

Table 7: China zinc trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Concentrate Imports (Gross)	2,969	3,183	3,822	3,648	4,135	4,723	4,097	5,331	2,762
Refined Zinc Imports	715	605	541	434	79	380	446	304	60
Unwrought Zinc and Alloys Imports	821	692	615	520	142	429	495	348	75
Refined Zinc Exports	22	62	29	5	81	9	17	94	21
Unwrought Zinc and Alloys Exports	24	64	32	7	83	12	22	101	21
Net Refined Imports	693	544	512	429	-2	371	429	210	39
YoY-%									
Concentrate Imports (Gross)		7.2%	20.1%	-4.6%	13.3%	14.2%	-13.3%	30.1%	9.0%
Refined Zinc Imports		-15.4%	-10.6%	-19.8%	-81.8%	380.1%	17.2%	-31.8%	-69.0%
Unwrought Zinc and Alloys Imports		-15.7%	-11.2%	-15.4%	-72.6%	201.7%	15.3%	-29.8%	-65.0%
Refined Zinc Exports		181.3%	-52.5%	-81.8%	1415.5%	-89.0%	88.2%	460.0%	67.8%
Unwrought Zinc and Alloys Exports		163.9%	-50.8%	-79.0%	1154.1%	-85.1%	78.4%	358.4%	35.3%

*2026 is through June.
Source: China Customs General Administration

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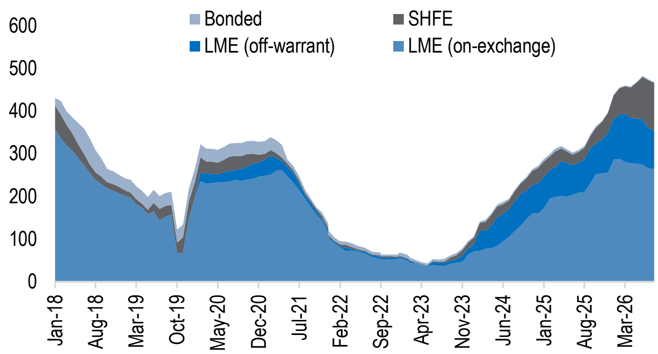
Nickel

Figure 91: Global visible nickel inventory

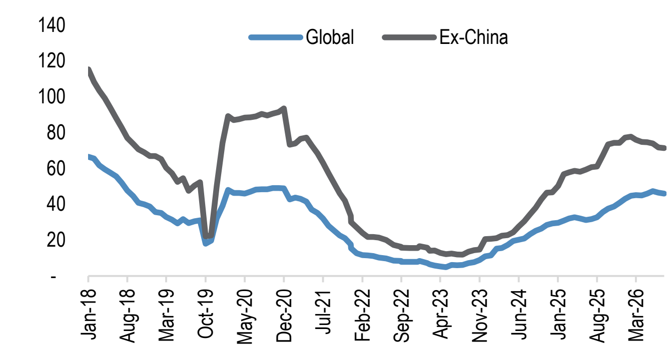
Thousand mt



Source: LME, SHFE, COMEX. Bonded warehouses are in China. As of 3rd August 2026.

Figure 92: Global visible nickel inventory in days of use

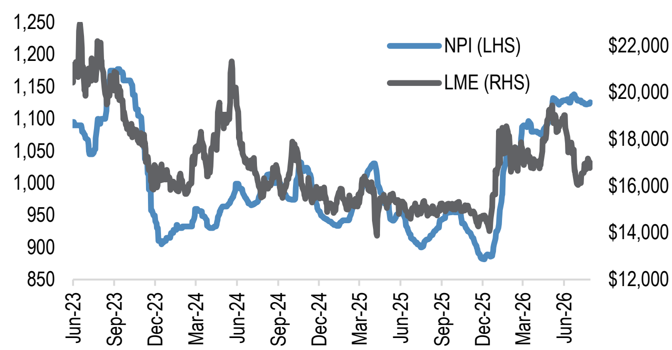
Days



Source: SHFE, J.P. Morgan Commodities Research

Figure 93: Chinese nickel pig iron prices (8-12%) and LME cash nickel prices

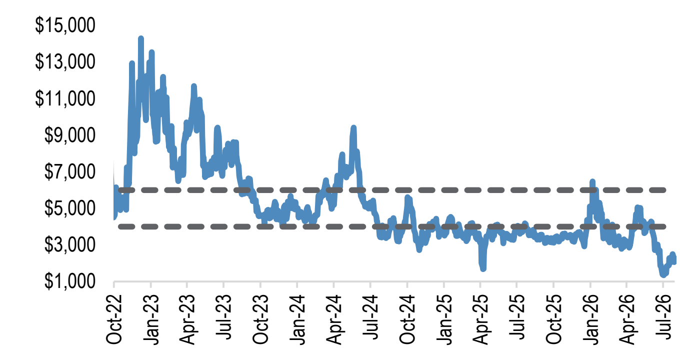
LHS: CNY/nickel tonne unit; RHS: US$/mt



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 29th July 2026.

Figure 94: Discount spread of NPI prices compared to LME cash nickel prices

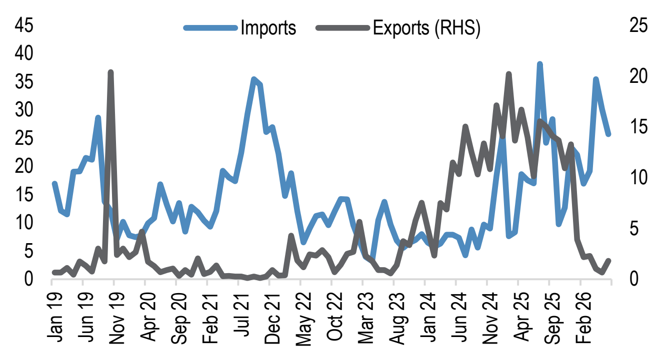
US$/mt. Dashed lines indicates level of estimated conversion costs



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 29th July 2026.

Figure 95: China monthly refined nickel imports and exports

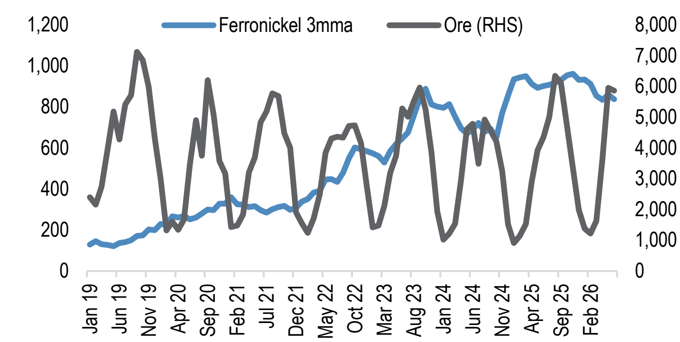
Thousand mt



Source: China Customs, Bloomberg Finance L.P.

Figure 96: China monthly ferronickel and nickel ore imports

Thousand mt



Source: China Customs, Bloomberg Finance L.P.

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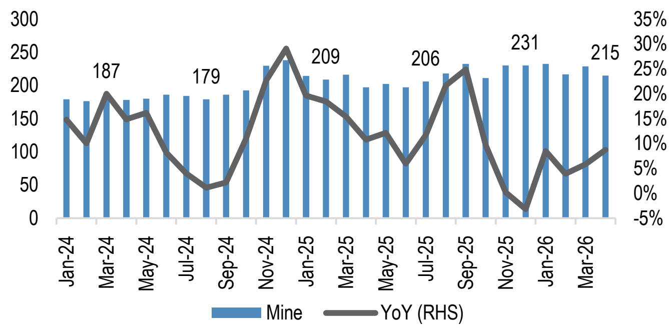
Figure 97: Indonesia monthly nickel mine production

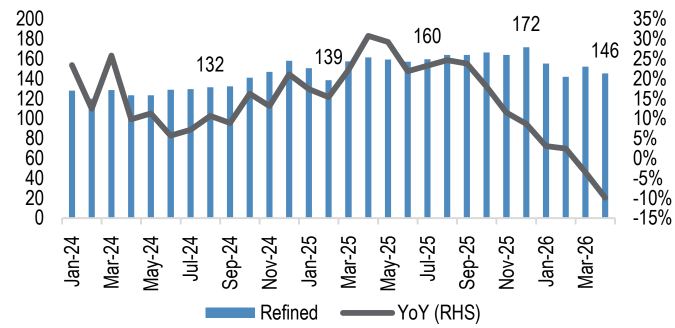
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 98: Indonesia monthly refined nickel production

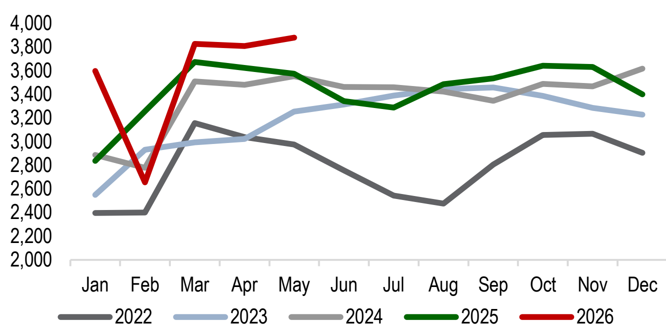
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 99: China stainless steel production

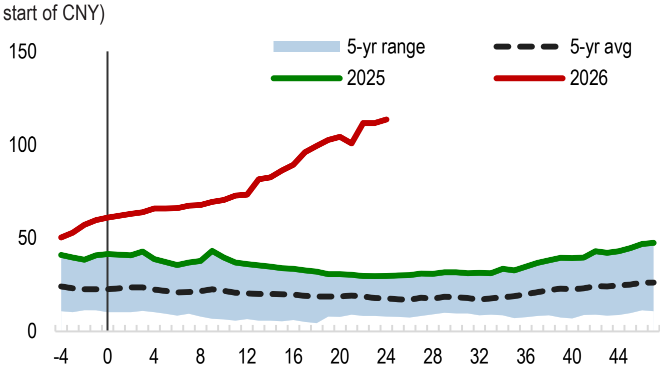
Thousand mt



Source: CRU

Figure 100: Total visible China nickel inventory (SHFE + Regional Warehouses)

Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

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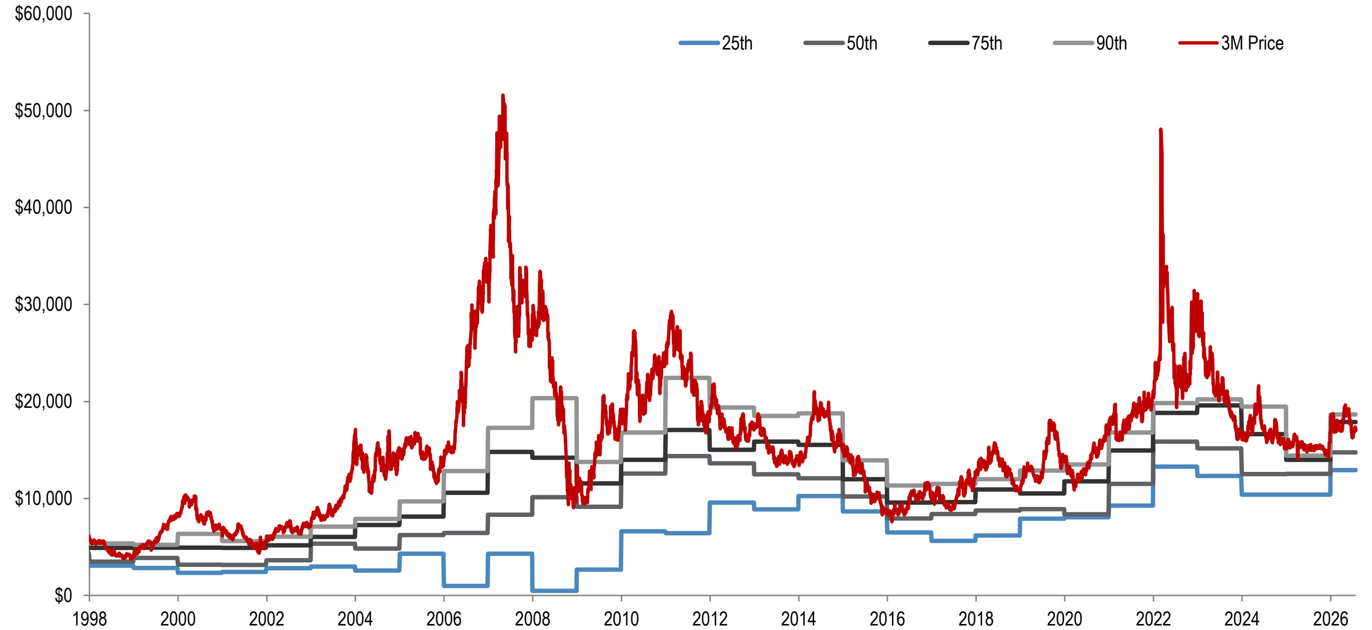
Figure 101: Global nickel industry C1 costs by %-tile and LME 3M nickel price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 29th July 2026.

Table 8: China nickel trade flow

Thousand mt; Percent Change, yoy

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Nickel Ore Imports	46,971	56,116	39,104	43,529	40,241	44,819	38,096	42,263	19,724
Refined Nickel Imports	212	193	131	261	156	91	97	231	149
Refined Nickel Exports	13	37	19	5	23	37	123	171	12
Net Refined Imports	199	156	112	256	133	54	-25	60	138
YoY-%									
Nickel Ore Imports		19.5%	-30.3%	11.3%	-7.6%	11.4%	-15.0%	10.9%	33.4%
Refined Nickel Imports		-8.9%	-32.3%	99.8%	-40.3%	-41.6%	6.8%	137.7%	57.9%
Refined Nickel Exports		182.5%	-49.3%	-71.3%	314.2%	62.7%	235.0%	39.7%	-86.6%

*2026 is through June.
Source: China Customs General Administration

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JPM metals price forecasts

Quarterly and annual averages

	4Q25A	1Q26A	2Q26A	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2025	2026	2027
Base Metals												
Aluminum	2,828	3,198	3,562	3,800	3,700	3,450	3,200	2,900	2,750	2,631	3,565	3,075
Copper	11,105	12,824	13,352	14,500	14,800	14,000	13,800	13,800	13,600	9,947	13,869	13,800
Nickel	14,892	17,338	18,090	17,000	17,250	17,500	18,000	18,000	17,500	15,164	17,419	17,750
Zinc	3,165	3,237	3,464	3,500	3,400	3,300	3,200	3,050	3,000	2,867	3,400	3,138
Precious Metals												
Gold	4,152	4,873	4,508	4,300	4,500	4,600	4,700	4,800	5,000	3,440	4,545	4,775
Silver	55.2	83.7	73.1	62.5	63.0	63.0	63.5	64.0	65.0	40.1	70.6	63.9
Platinum	1,694	2,204	1,916	1,700	1,800	1,875	1,900	1,950	1,950	1,282	1,905	1,919
Palladium	1,478	1,706	1,409	1,300	1,350	1,400	1,350	1,250	1,200	1,151	1,441	1,300

Source: J.P. Morgan Commodities Research. *Base metals are LME cash and precious metals are spot.

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